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SCHOOL OF COMPUTING AND INFORMATION TECHNOLOGY



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OmniPlant - Manufacturing ERP - Capstone 2

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Nguyen Phuc Hau - 2231200065
Dang Cao Cuong - 2231200087

Software Engineering

Ho Chi Minh City, August, 2026

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ABSTRACT

OmniPlant is a web-based Manufacturing Enterprise Resource Planning system designed for small and medium-sized assembly manufacturers. The Capstone 2 phase transforms the analytical foundation and prototype produced in Capstone 1 into an API-backed production loop that links customer demand, material planning, production execution, quality disposition, inventory and traceability. The implemented flow begins with a confirmed Sales Order, creates an immutable Material Requirements Planning run, explodes active multi-level Bills of Materials, nets eligible stock and safety stock, emits MAKE and deferred BUY suggestions, converts approved production suggestions into Work Orders, reserves and issues materials, records manual work-in-progress results, approves production receipts and releases accepted finished goods through a HOLD-to-AVAILABLE quality gate.

The solution combines a Next.js and TypeScript frontend with a Spring Boot modular-monolith backend, PostgreSQL, Redis, JWT authentication and dynamic role-based access control. Capstone 2 emphasizes operational correctness: immutable BOM and Routing snapshots, Plant and Warehouse scoping, lot/serial genealogy, exclusion of HOLD and REJECTED stock from planning, retry-safe idempotent commands, confirmation for integrity-sensitive actions, and bilingual desktop/tablet workspaces. Live integration and regression acceptance demonstrate the complete production spine from Sales Order demand to quality-approved inventory. Full Purchasing, delivery and invoicing, detailed manufacturing costing, OEE and MES/PLC integration remain intentionally deferred to Capstone 3. Standalone Production Estimation is not part of the focused Capstone 2 MVP because the authoritative Planning Run covers the operational MRP purpose.

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We are grateful to the faculty members of the School of Computing and Information Technology at Eastern International University for the technical foundation, facilities and academic environment provided throughout the project.

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TABLE OF CONTENTS

ABSTRACT	I
ACKNOWLEDGEMENT	II
TABLE OF CONTENTS.....	III
LIST OF FIGURES.....	VII
LIST OF TABLES.....	IX
LIST OF ABBREVIATIONS.....	X
CHAPTER 1. INTRODUCTION	1
1.1. Project Background	1
1.2. Problem Statement	1
1.3. Project Objectives.....	2
1.4. Project Scope.....	2
1.4.1. In-Scope for Capstone 2	2
1.4.2. Out-of-Scope and Deferred	3
1.5. Capstone 2 Contributions	3
1.6. Report Structure	4
CHAPTER 2. RELATED WORKS.....	5
2.1. Overview of Existing Systems	5
2.2. Comparative Analysis	5
2.3. Identified Gap and Design Position.....	5
CHAPTER 3. SYSTEM ARCHITECTURE, ANALYSIS, DESIGN AND IMPLEMENTATION	7
3.1. Chapter Overview.....	7
3.2. System Architecture	7
3.2.1. Architectural Style.....	7
3.2.2. Technology Stack	8
3.3. Functional Analysis and Actor Model.....	9
3.3.1. System Actors.....	9
3.3.2. Use Case Model.....	10
3.4. End-to-End Business Process Design.....	11

3.4.1. System Setup and Access Control.....	13
3.4.2. Sales and Material Planning	13
3.4.3. Purchasing Flow	14
3.4.4. Manufacturing Execution Flow	14
3.4.5. Receipt, Quality, and Inventory Update	14
3.5. Database Design	15
3.5.1. Data Modeling Principles	15
3.5.2. System Administration and Integration ERD.....	15
3.5.3. Product and Production Master Data ERD.....	17
3.5.4. Sales and Planning ERD.....	18
3.5.5. Purchasing ERD	19
3.5.6. Inventory and Warehouse ERD.....	20
3.5.7. Manufacturing Execution and Quality ERD	21
3.5.8. Cross-Module Data Relationships.....	23
3.6. Business Rules and State Controls	23
3.7. API, Security, and Transaction Resilience.....	24
3.8. Frontend Interaction Design.....	25
3.9. System Operation and Deployment.....	25
3.10. Chapter Summary	26
CHAPTER 4. RESULTS AND DISCUSSION	27
4.1. Implementation Results	27
4.2. API-Backed Interface Evidence	27
4.2.1. Dashboard Overview.....	28
4.2.2. Item Master.....	29
4.2.3. Stock Balance	30
4.2.4. Inventory Lots	31
4.2.5. Stock Movement History.....	32
4.2.6. BOM List.....	33
4.2.7. Active BOM Detail	34
4.2.8. Multi-Level BOM Tree	35
4.2.9. Sales Order List	36
4.2.10. Sales Order Detail	37
4.2.11. Material Planning Workspace	38

4.2.12. Planning Run Detail	39
4.2.13. Work Order List	40
4.2.14. Work Order Detail.....	41
4.2.15. Material Issue Workspace	42
4.2.16. Production Execution Workspace	43
4.2.17. Production Receipt List.....	44
4.2.18. Inventory Lots after Quality Control.....	45
4.2.19. Login and Session Initialization	46
4.2.20. User Management.....	47
4.2.21. Role Management.....	48
4.2.22. Permission Catalog.....	49
4.2.23. Access Scope Management	50
4.2.24. User Assignment	51
4.2.25. Organization Structure	52
4.2.26. Unit of Measure Management.....	53
4.2.27. Audit Logs.....	54
4.3. Validation and Acceptance Results.....	54
4.4. Operational Demonstration Result	55
4.5. Engineering Discussion.....	55
4.5.1. Planning Run versus Standalone Production Estimation	55
4.5.2. Immutable Snapshots and Idempotency.....	55
4.5.3. Availability and Quality Semantics.....	56
4.5.4. ERP Execution without MES.....	56
4.6. Known Limitations.....	56
CHAPTER 5. CONCLUSION AND FUTURE WORKS.....	57
5.1. Conclusion.....	57
5.2. Future Works.....	57
5.3. Recommended Capstone 3 Sequence.....	58
5.4. Operational State Reference.....	58
REFERENCES	59
APPENDIX	60
A.1. Focused MVP Screen Inventory.....	60
A.2. End-to-End Demonstration Checklist.....	60

LIST OF FIGURES

Figure 1. OmniPlant system architecture and major technical components.....	8
Figure 2. Use case diagram of OmniPlant actors and major system functions.	10
Figure 3. Swimlane activity diagram of the end-to-end OmniPlant business process.	13
Figure 4. ERD of the System Administration and Integration module.	16
Figure 5. ERD of the Product and Production Master Data module.	17
Figure 6. ERD of the Sales and Planning module.	18
Figure 7. ERD of the Purchasing module.....	19
Figure 8. ERD of the Inventory and Warehouse module.	20
Figure 9. ERD of the Manufacturing Execution and Quality module.....	23
Figure 10. Dashboard with API-backed operational overview and alerts	28
Figure 11. Item Master with item types, UOM and tracking-policy evidence	29
Figure 12. Stock Balance with on-hand, reserved and available quantities	30
Figure 13. Inventory Lots with AVAILABLE, HOLD and REJECTED status evidence	31
Figure 14. Stock Movement history with transaction types and trace references	32
Figure 15. BOM list with active and draft lifecycle visibility.....	33
Figure 16. Active BOM detail with revision, output quantity and component lines.....	34
Figure 17. Multi-level BOM tree showing nested sub-assemblies and component quantities.....	35
Figure 18. Sales Order list with lifecycle and fulfillment status evidence	36
Figure 19. Sales Order detail with line allocation and fulfillment quantities.....	37
Figure 20. Material Planning workspace with confirmed demand and Planning Run history	38
Figure 21. Planning Run detail with shortage and MAKE/BUY supply decisions.....	39
Figure 22. Work Order list with production lifecycle status evidence	40
Figure 23. Work Order detail with complete reservation readiness	41
Figure 24. Material Issue workspace with component and issue-history evidence.....	42
Figure 25. Production Execution workspace with good, scrap and rework quantities.....	43
Figure 26. Production Receipt list with approval and QC lifecycle states	44
Figure 27. Inventory Lots after QC with AVAILABLE, HOLD and REJECTED output.....	45
Figure 28. Login interface for session and Plant-scope initialization	46
Figure 29. User management with account lifecycle status	47
Figure 30. Role management with active responsibility groups	48
Figure 31. Permission catalog with resource and action codes	49
Figure 32. Access Scope management across organization levels.....	50
Figure 33. User Assignment form linking User, Role and Scope	51
Figure 34. Organization structure with Company, Plant and Warehouse evidence	52
Figure 35. UOM management with reusable measurement codes	53
Figure 36. Audit Logs with actor, entity, action and timestamp evidence	54

LIST OF TABLES

Table 1. Capstone 2 functional scope.....	3
Table 2. Comparison of related manufacturing ERP systems.....	5
Table 3. Technology stack.....	9
Table 4. Actor responsibilities.....	9
Table 5. Cross-module data relationships.	23
Table 6. Core business validation rules.	24
Table 7. Operational startup and verification checklist.	26
Table 8. Capstone 2 delivery summary	27
Table 9. Quality and acceptance evidence	55
Table 10. Recommended Capstone 3 implementation sequence	58
Table 11. Primary document state transitions	58
Table 12. Focused Capstone 2 MVP navigation	60

LIST OF ABBREVIATIONS

No.	Term	Meaning
1	API	Application Programming Interface
2	BOM	Bill of Materials
3	ERP	Enterprise Resource Planning
4	FEFO	First Expired, First Out
5	FG	Finished Good
6	JWT	JSON Web Token
7	MES	Manufacturing Execution System
8	MPS	Master Production Schedule
9	MRP	Material Requirements Planning
10	OEE	Overall Equipment Effectiveness
11	QC	Quality Control
12	RBAC	Role-Based Access Control
13	ROP	Reorder Point
14	SME	Small and Medium-sized Enterprise
15	SO	Sales Order
16	UOM	Unit of Measure
17	WIP	Work in Progress
18	WO	Work Order

CHAPTER 1. INTRODUCTION

1.1. Project Background

Manufacturing SMEs operate through a tightly coupled flow of customer demand, material availability, engineering definitions and shop-floor execution. A finished product may depend on multiple levels of semi-finished assemblies and raw materials, while the same warehouse stock can already be reserved, quarantined or rejected. When these relationships are managed with spreadsheets or disconnected applications, a production decision can be based on stale quantities, obsolete BOM versions or incomplete traceability.

Capstone 1 established the OmniPlant concept, domain analysis, database direction, role model and enterprise-style interface. Capstone 2 moves beyond a prototype by implementing the production spine as an integrated ERP workflow. Customer demand is captured as a Sales Order, translated into an immutable Planning Run, converted into traceable Work Orders and executed through controlled stock and quality transactions. The system deliberately avoids claiming MES behavior: actual progress, good quantity, scrap, rework and time are entered manually by authorized users.

The frontend is designed for desktop and tablet use by administrators, production managers, warehouse staff and shop-floor operators. Its visual approach is inspired by ERPNext: compact list views, clear filters, stable forms, status badges and confirmation dialogs are prioritized over promotional or consumer-style layouts.

1.2. Problem Statement

The implementation addresses five operational gaps that remain common in smaller manufacturing organizations:

- **Disconnected demand and production:** Confirmed customer orders are not consistently converted into material requirements and executable production documents.
- **Unreliable availability:** Physical on-hand stock is often confused with stock that is actually available after reservations and quality status are considered.
- **Version drift:** A Work Order can become inconsistent when its BOM or Routing master changes after planning unless the selected versions are snapshotted.
- **Weak execution traceability:** Material issues, WIP confirmations, output receipts and lot genealogy are difficult to reconcile when each step is recorded separately.
- **Unsafe retries and approvals:** Network failures, duplicate submissions and unconfirmed stock-changing actions can create duplicate or inconsistent records.

1.3. Project Objectives

- **Deliver an end-to-end production loop:** Implement the sequence from confirmed Sales Order demand to QC-approved finished-goods inventory.
- **Provide authoritative MRP:** Use immutable Planning Runs, multi-level BOM explosion, safety-stock netting and explicit MAKE/BUY suggestions.
- **Protect inventory integrity:** Separate on-hand, reserved and available quantities; exclude HOLD and REJECTED lots; use append-only movement records.
- **Preserve production genealogy:** Link Sales Order lines, Work Orders, BOM/Routing snapshots, issued lots, WIP confirmations, Production Receipts and output lots.
- **Enforce role and scope control:** Apply dynamic permissions and Plant/Warehouse access at navigation, route, action and data levels.
- **Support resilient ERP operation:** Use idempotency keys, stable API envelopes, explicit errors and forms that preserve user input after failures.
- **Validate the implementation:** Cover core rules through automated regression, API contract checks and desktop/tablet acceptance.

1.4. Project Scope

1.4.1. In-Scope for Capstone 2

Area	Implemented Capstone 2 boundary
Platform and administration	Authentication, rotating session tokens, users, roles, permissions, access scopes, Plant selection and audit visibility.
Manufacturing foundation	UOM, Item Master, Item-Warehouse settings, multi-level BOM, inventory balances, lots and stock movements.
Demand and planning	Sales Orders, confirmed planning demand, immutable Planning Runs, BOM explosion, safety-stock netting, exception reporting and MAKE/BUY suggestions.
Production documents	Work Order creation/conversion, demand allocation, BOM/Routing snapshots, material readiness, release and cancellation.
Execution	Reservation, Material Issue, over-BOM variance seam, manual good/scrap/rework/time confirmation and WIP trace.
Receipt and quality	Partial Production Receipts, Manager approval/rejection, HOLD output and QC disposition to AVAILABLE or REJECTED.

Area	Implemented Capstone 2 boundary
Reporting and UX	Inventory Dashboard, low-stock alerts, movements, bilingual EN/VI UI, loading/empty/error states and desktop/tablet support.

Table 1. Capstone 2 functional scope

1.4.2. Out-of-Scope and Deferred

- **Full Purchasing:** Supplier lifecycle, Purchase Requisition, Purchase Order, Goods Receipt and commercial approval workflow are deferred to Capstone 3. BUY suggestions remain visible but read-only in the focused MVP.
- **Sales fulfillment:** Delivery orders, invoicing, payment and customer credit are deferred.
- **MES and telemetry:** PLC/IoT integration, automatic machine signals, dispatching, downtime and OEE are deferred; Capstone 2 uses manual ERP confirmation.
- **Detailed costing:** Labor, overhead, WIP valuation, standard-versus-actual cost and profitability are deferred.
- **Advanced scheduling:** Capacity warnings and Routing context are retained, but automatic finite-schedule optimization is not part of the focused MVP.
- **Standalone Production Estimation:** The legacy what-if calculator is not included in the Capstone 2 production spine because the API-backed Planning Run is the authoritative MRP function.

1.5. Capstone 2 Contributions

The principal contribution of Capstone 2 is not a collection of unrelated screens but a controlled state transition across business documents. Each transition has an owner, permission, validation rule, inventory effect and trace relationship. This design allows the same flow to be demonstrated from demand through approved output without inventing MES data or bypassing the inventory ledger.

A second contribution is the separation of master data from execution evidence. Active BOM and Routing versions are selected at planning time, while a Work Order receives immutable snapshots. The resulting document can therefore be audited even when engineering masters later change.

1.6. Report Structure

- Chapter 1 introduces the implementation problem, objectives, Capstone 2 scope and contributions.
- Chapter 2 compares the focused OmniPlant approach with established ERP manufacturing systems.
- Chapter 3 documents architecture, actors, modules, workflows, business rules, APIs, security, traceability and system operation.
- Chapter 4 presents implemented results, API-backed interface evidence, testing outcomes, limitations and engineering discussion.
- Chapter 5 concludes the Capstone 2 phase and defines the Capstone 3 roadmap.

CHAPTER 2. RELATED WORKS

2.1. Overview of Existing Systems

Commercial and open-source ERP platforms provide useful references for manufacturing workflows. SAP S/4HANA supports large-scale material requirements planning, production planning and detailed scheduling. Odoo provides Bills of Materials, manufacturing orders, work orders, work centers and shop-floor-oriented execution. ERPNext provides production plans, Work Orders, Stock Entries, Job Cards, BOM explosion and inventory integration in a comparatively accessible open-source suite.

These systems confirm the standard manufacturing pattern adopted by OmniPlant: engineering masters precede planning; confirmed demand is converted into planned supply; Work Orders define internal production; material consumption and finished output update inventory through controlled documents. OmniPlant intentionally follows these recognizable concepts so operators do not need to learn an unconventional model.

2.2. Comparative Analysis

System	Relevant strengths	Capstone 2 consideration
SAP S/4HANA	Enterprise-scale MRP, PP/DS, costing and global integration	High implementation cost and complexity for a focused SME capstone deployment
Odoo Manufacturing	Configurable manufacturing orders, work centers, shop-floor and inventory integration	Broad suite and configuration surface can exceed the targeted workflow
ERPNext	Accessible BOM, Production Plan, Work Order and stock-oriented workflow	Still includes a wider generalized ERP model requiring adaptation
OmniPlant	Focused demand-to-QC production loop, dynamic RBAC, lot genealogy and explicit manual-without-MES boundary	Narrower scope; purchasing, finance, MES and detailed costing are deferred

Table 2. Comparison of related manufacturing ERP systems

2.3. Identified Gap and Design Position

The gap addressed by OmniPlant is not the absence of mature ERP software. It is the difficulty of applying a large generalized platform to a small manufacturing team that needs a

transparent, demonstrable and traceable production loop. OmniPlant therefore selects a smaller number of concepts and makes their rules explicit: what stock is eligible, when a Work Order is releasable, who approves a receipt, how quality affects availability and how one can navigate from a customer order to an output lot.

The project also distinguishes operational planning from ad-hoc estimation. ERPNext and comparable suites demonstrate that production planning begins with demand and creates executable production documents. Capstone 2 adopts this operational model through immutable Planning Runs. A separate one-product what-if estimator may still be useful in a future presentation or quotation context, but it is not allowed to replace the MRP evidence in the implemented scope.

CHAPTER 3. SYSTEM ARCHITECTURE, ANALYSIS, DESIGN AND IMPLEMENTATION

3.1. Chapter Overview

This chapter presents the analysis, architectural design, business-process design, database structure, and implementation principles of OmniPlant. The chapter begins with the overall technical architecture and actor model before describing how the main actors interact with the system. It then explains the end-to-end operational workflow through a swimlane activity diagram.

The database design is divided into six major functional modules. This modular presentation makes the data model easier to understand while preserving the relationships among administration, master data, sales and planning, purchasing, inventory, and manufacturing execution. The final sections describe the core business rules, API and security design, frontend interaction principles, and deployment environment.

3.2. System Architecture

3.2.1. Architectural Style

OmniPlant is implemented as a web-based Manufacturing ERP system using a modular-monolith architecture. This approach maintains a single deployable backend while separating the application into clear functional modules. It provides simpler deployment and transaction management than a distributed microservice architecture while still preserving domain boundaries.

The frontend is responsible for user interaction, client-side validation, permission-aware navigation, query state, internationalization, and responsive presentation. The backend remains the authoritative source for authentication, authorization, business-state transitions, inventory effects, material planning calculations, approval rules, and database consistency.

PostgreSQL stores transactional and master data, while Redis supports session and token-management functions. Communication between the frontend and backend is performed through RESTful API endpoints under the `/api/v1` namespace.

OmniPlant Capstone 2 - System Architecture

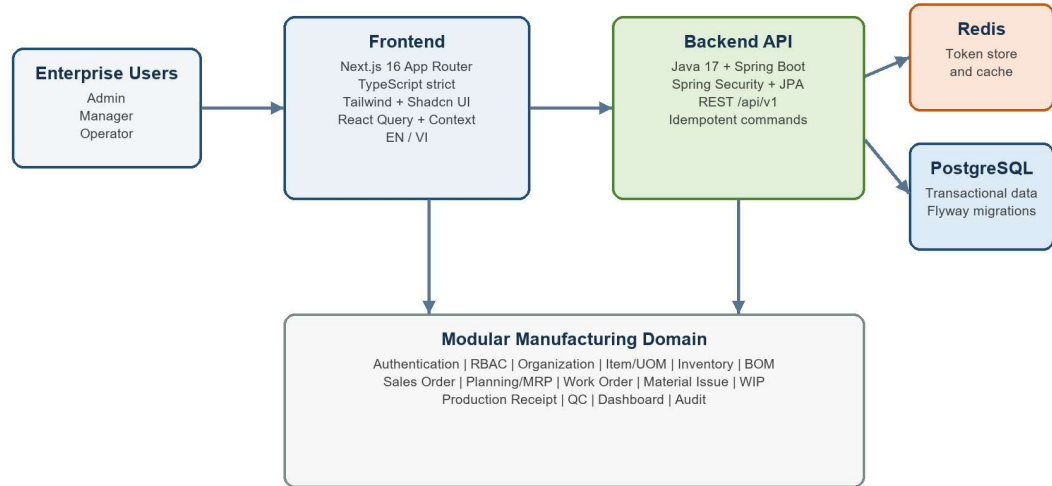


Figure 1. OmniPlant system architecture and major technical components.

The architecture separates presentation concerns from business authority. Although the frontend prevents invalid actions where possible, all critical rules are revalidated by the backend. This includes Plant and Warehouse access, Work Order release readiness, material availability, approval permissions, quality disposition, and idempotent inventory transactions.

3.2.2. Technology Stack

The major technologies and their responsibilities are summarized in the following table.

Layer	Technology and responsibility
Frontend	Next.js App Router, React, and strict TypeScript.
UI and styling	Tailwind CSS, customized Shadcn UI, and Lucide icons.
Client state	React Query for server state and Context API for authentication and selected Plant.
Internationalization	next-intl with English and Vietnamese namespaces.
Backend	Java 17, Spring Boot, Spring Security, and JPA/Hibernate.
Persistence	PostgreSQL with Flyway migrations and Redis for session support.

Integration	RESTful /api/v1 endpoints, UUID identifiers, and a standard response envelope.
-------------	--

Table 3. Technology stack.

3.3. Functional Analysis and Actor Model

3.3.1. System Actors

OmniPlant defines three primary human actors: Administrator, Manager, and Operator. The actors are separated according to responsibility and authority rather than only by screen visibility.

Actor	Main responsibilities
Administrator	Maintains users, roles, permissions, access scopes, organizational structure, and system-level master data. Reviews audit information and controls access assignments.
Manager	Manages Sales Orders, material planning, supply decisions, Work Orders, purchasing documents, production approvals, quality disposition, and operational monitoring.
Operator	Performs authorized warehouse and production activities, including reservation, material issue, production reporting, goods receipt, and Production Receipt submission.
System	Performs authentication, authorization, MRP calculation, BOM explosion, inventory updates, status transitions, variance calculation, and audit logging.

Table 4. Actor responsibilities.

The System is not a human actor. It represents automated application behavior triggered by the actions of authenticated users.

3.3.2. Use Case Model

The following use case diagram presents the functional boundary of OmniPlant and the interactions between its three human actors and the major system functions.

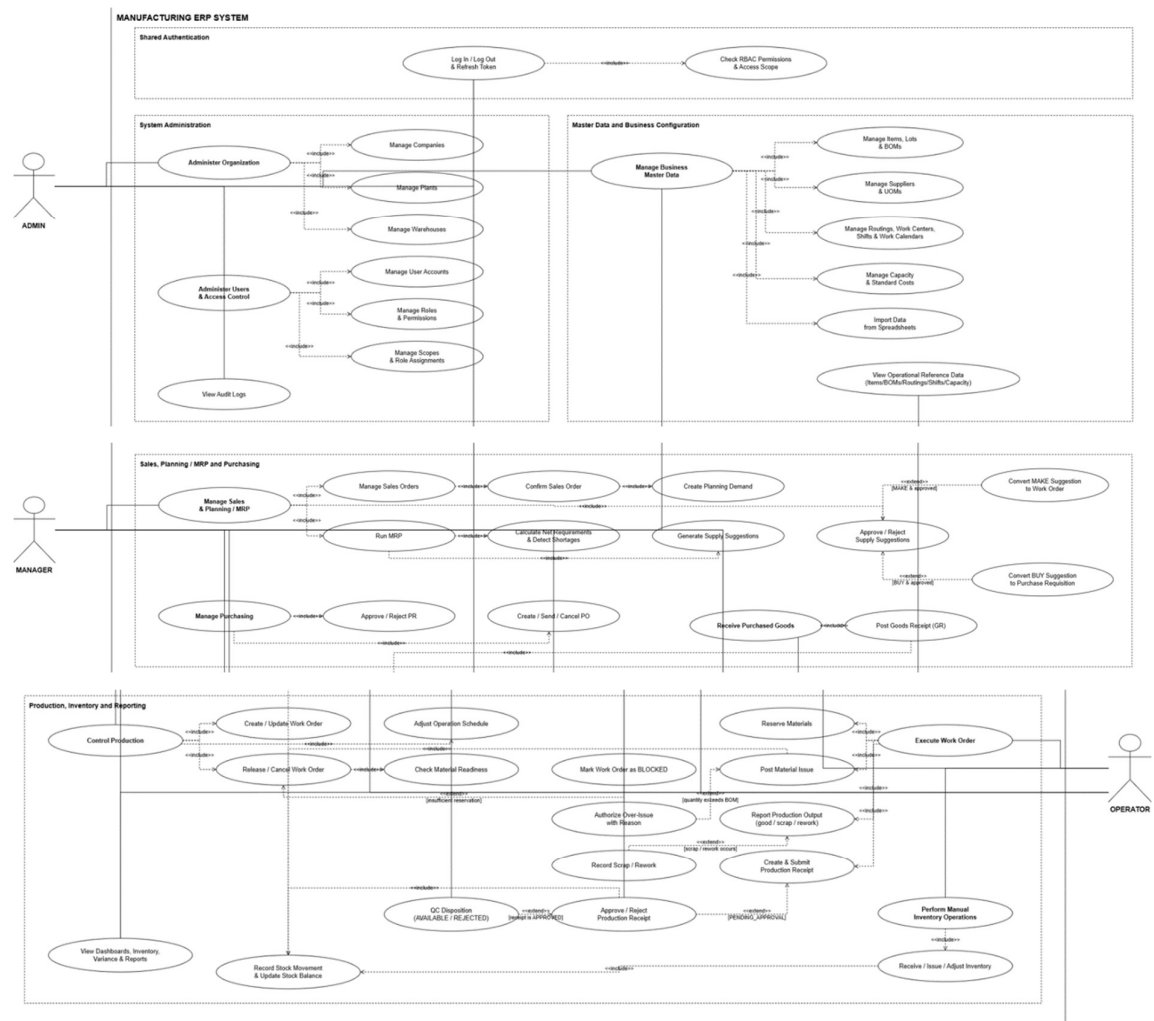


Figure 2. Use case diagram of OmniPlant actors and major system functions.

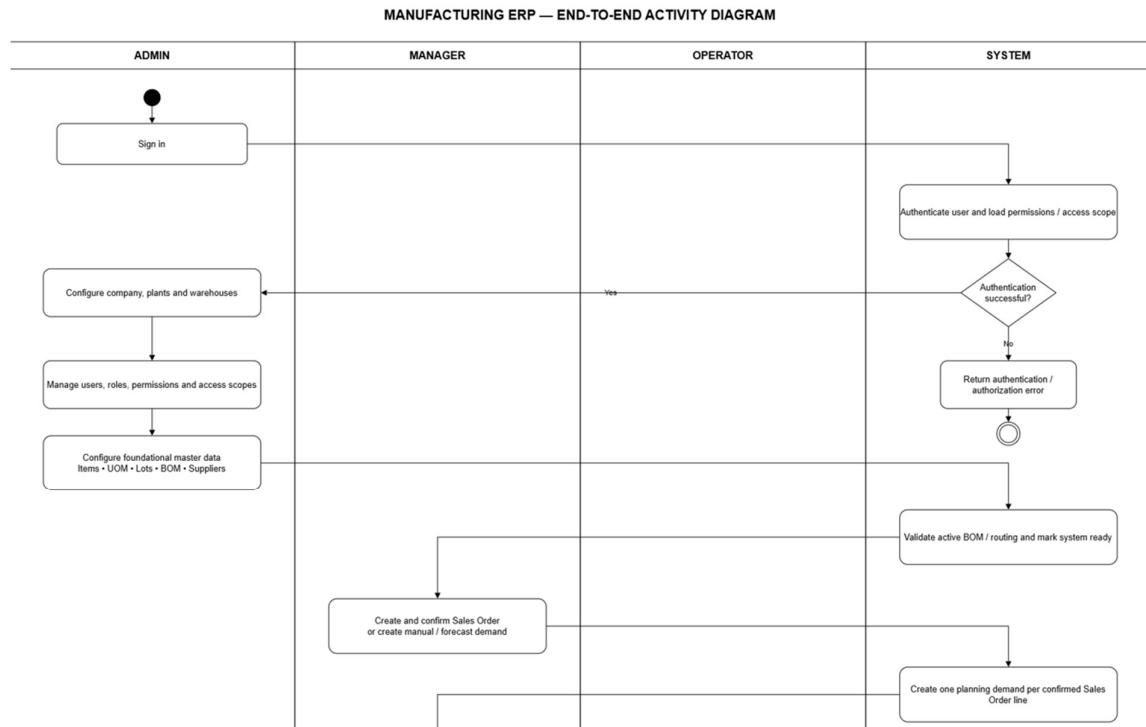
The use case diagram describes functional responsibility but does not represent execution order. Administrator use cases focus on system configuration, organizational data, access control, and auditability. Manager use cases cover planning, approval, production control, purchasing, and quality decisions. Operator use cases cover the direct execution of inventory, receiving, and manufacturing activities.

Several use cases are shared between actors but are controlled by different permissions. For example, both Managers and Operators may view Work Orders, while only authorized Managers can create or release them. Operators may create and submit Production Receipts, but they cannot approve their own receipts or perform final quality disposition. This

separation of duties prevents the same user from executing and approving an integrity-sensitive transaction.

3.4. End-to-End Business Process Design

The operational behavior of OmniPlant is document-driven. A user cannot move directly from demand to inventory output without passing through the required planning, production, approval, and quality-control documents. The following workflow is divided into four swimlanes: Administrator, Manager, Operator, and System.



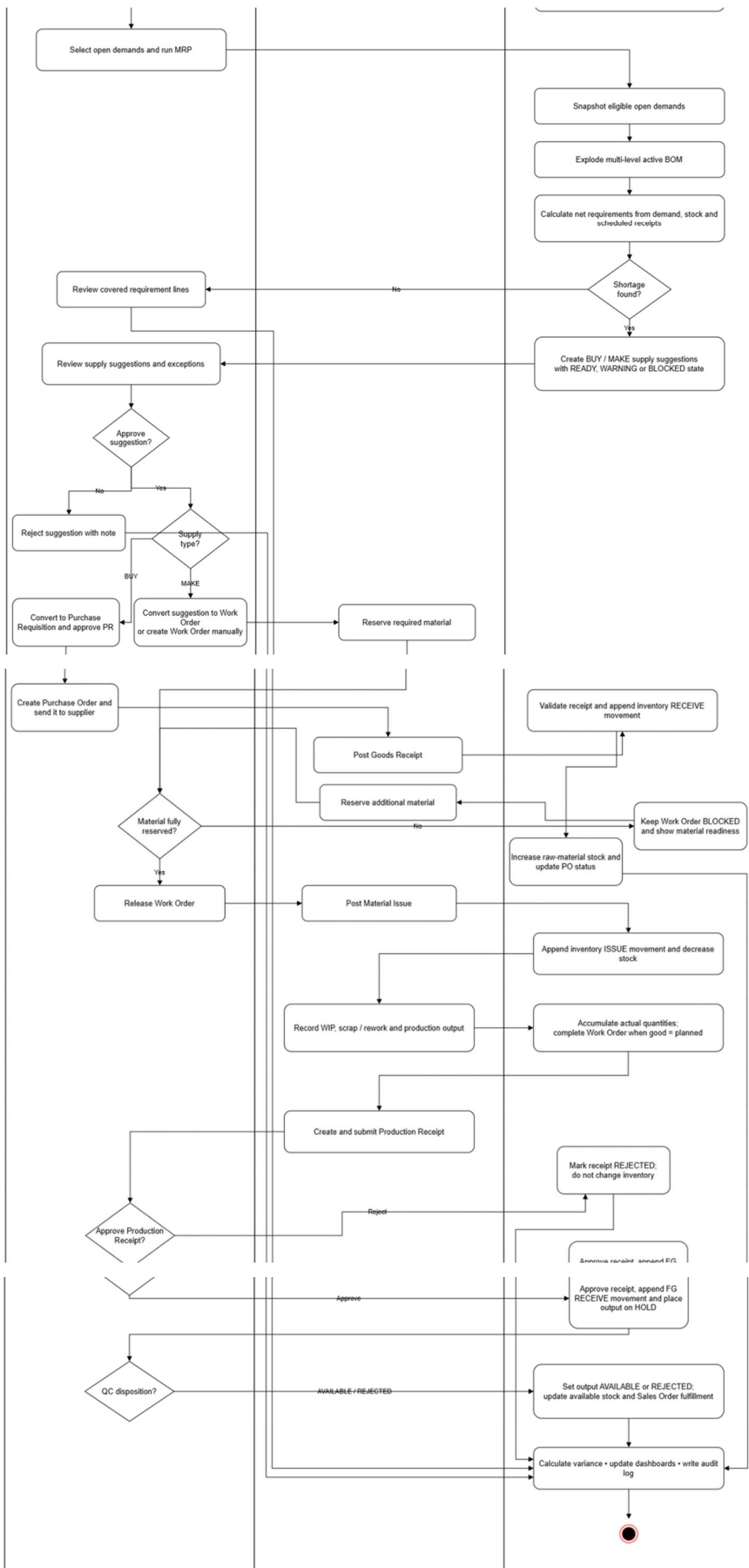


Figure 3. Swimlane activity diagram of the end-to-end OmniPlant business process.

The swimlane diagram identifies both execution sequence and responsibility. Human decisions are assigned to the role that owns the business outcome, while validation, calculation, inventory posting, and audit recording are placed in the System lane. The diagram also demonstrates that the Administrator prepares the operating environment but does not participate directly in daily purchasing or production execution.

3.4.1. System Setup and Access Control

The process begins with authentication and authorization. The System validates the user credentials and loads the corresponding roles, permissions, and access scopes. If authentication fails, the request is rejected without entering the business workflow.

Before operational activities begin, the Administrator configures the Company, Plants, Warehouses, users, roles, permissions, and access scopes. Foundational master data such as units of measure, Items, Bills of Materials, suppliers, Routings, Work Centers, shifts, and work calendars must also be available according to the relevant module permissions.

The System validates important master-data invariants. Examples include preventing circular BOM structures, requiring an active BOM for material explosion, and ensuring that production master data belongs to the correct Company or Plant context.

3.4.2. Sales and Material Planning

A Manager creates and confirms a Sales Order or records a manual or forecast demand. Confirmation converts the commercial requirement into an eligible planning demand. Draft Sales Orders do not participate in MRP.

The Manager selects eligible demand and starts an MRP run. The System creates an immutable snapshot of the selected demand, explodes the active multi-level BOM, evaluates eligible inventory and scheduled supply, and calculates the net requirement.

$$netRequirement = \max(0, grossRequirement + stockTarget - projectedAvailable)$$

If sufficient supply exists, the requirement is marked as covered. Otherwise, the System creates a supply suggestion. A suggestion may recommend purchasing the material or manufacturing it internally. The Manager reviews the suggestion, its exception state, and any missing BOM or Routing information before approving, rejecting, or converting it.

3.4.3. Purchasing Flow

For a BUY requirement, an approved supply suggestion may be converted into a Purchase Requisition. The Manager reviews and approves the requisition before converting it into a Purchase Order and sending the order to the selected supplier.

When the supplier delivers the material, the Operator posts a Goods Receipt. The System validates the Purchase Order line, Warehouse, received quantity, lot information, and idempotency key. A valid receipt creates an inventory RECEIVE movement, increases raw-material stock, and updates the received quantity and status of the Purchase Order.

3.4.4. Manufacturing Execution Flow

For a MAKE requirement, an approved suggestion is converted into a Work Order. The Work Order stores the selected product, planned quantity, output Warehouse, BOM evidence, Routing evidence, and demand allocations.

Materials are reserved before release. Reservation reduces available quantity but does not change physical on-hand stock. If the reservation does not cover all component requirements, the System keeps the Work Order in the BLOCKED state and reports the missing quantities. After sufficient material is reserved, the Manager releases the Work Order.

The Operator posts a Material Issue for eligible reserved components. The System creates an inventory ISSUE movement and decreases the corresponding stock balance. Issues above the Work Order BOM requirement require an explicit reason and an authorized override.

During production, the Operator records good, scrap, and rework quantities. These transactions update the Work Order actual quantities and preserve WIP evidence. OmniPlant records these results manually and does not claim automated MES or machine-data integration.

3.4.5. Receipt, Quality, and Inventory Update

After production output is reported, the Operator creates and submits a Production Receipt. Submission moves the receipt into the PENDING_APPROVAL state without immediately changing inventory.

An authorized Manager may approve or reject the receipt. Rejection requires a reason and has no stock effect. Approval creates a finished-goods RECEIVE movement and places the received output in HOLD status.

Quality disposition is performed separately from receipt approval. An AVAILABLE result releases the output for planning and Sales Order fulfillment. A REJECTED result keeps

the quantity unavailable for operational use. The System then updates inventory balances, Sales Order fulfillment, operational dashboards, variance information, and audit logs.

3.5. Database Design

3.5.1. Data Modeling Principles

The OmniPlant database uses UUID primary keys and explicit foreign keys for physical relationships. Transactional tables retain timestamps, actor identifiers, and version fields where required for auditability and optimistic concurrency.

The model separates reusable master data, lifecycle-controlled business documents, and append-only ledger or execution evidence. Some fields, such as `reference_id`, are intentionally polymorphic and therefore do not have a physical foreign key. These fields are documented as logical relationships rather than being presented as database-enforced relationships.

Each ERD is independently readable. A table marked as an external reference contains only the relationship key required by the module diagram; its complete definition is presented in the module that owns the table.

3.5.2. System Administration and Integration ERD

This module defines the organizational and authorization foundation of OmniPlant. It includes users, Companies, Plants, Warehouses, roles, permissions, access scopes, assignments, audit logs, and data-import records.

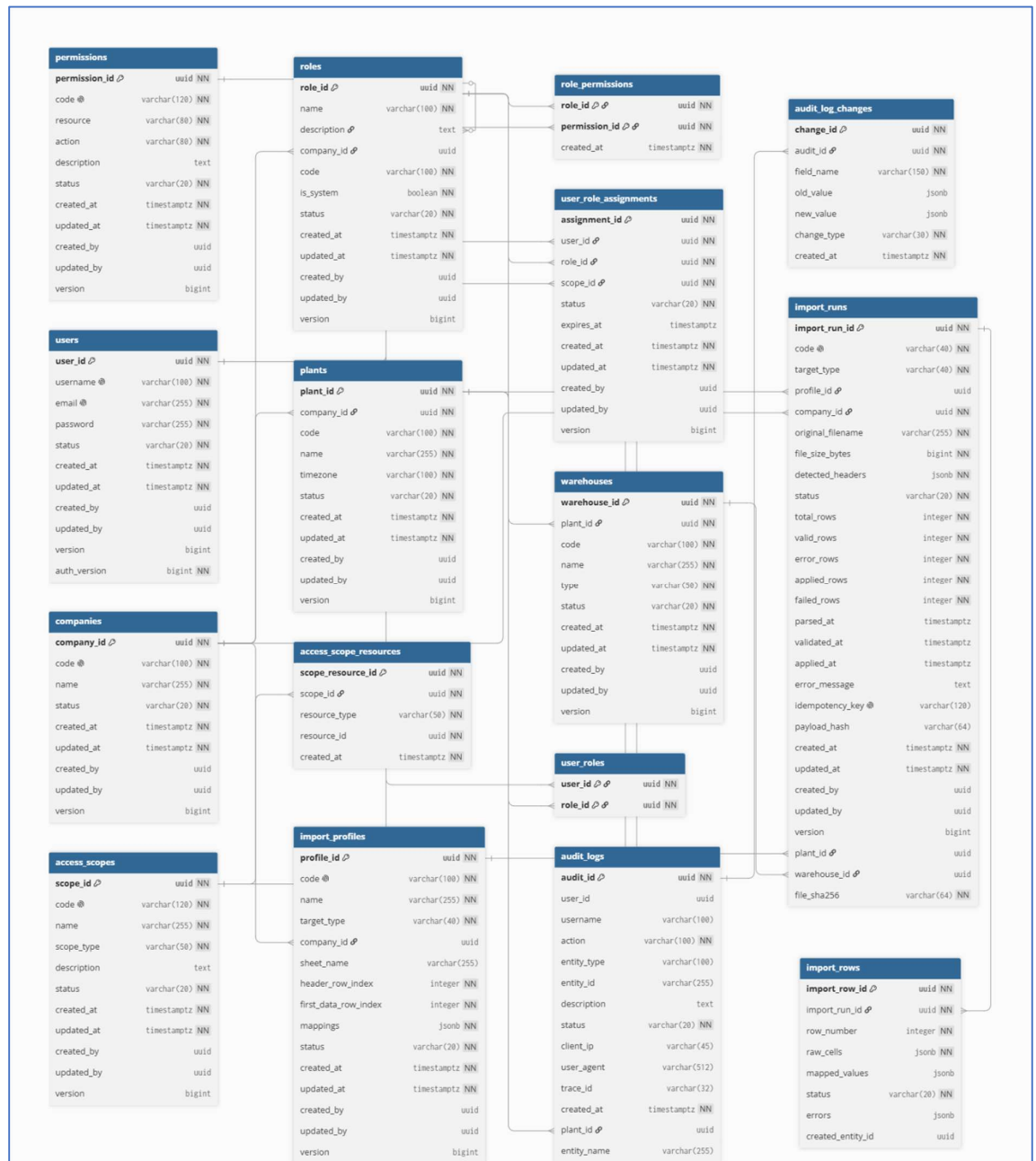


Figure 4. ERD of the System Administration and Integration module.

A Company contains multiple Plants, and each Plant contains multiple Warehouses. Roles are connected to permissions through role_permissions. A user receives operational authority through user_role_assignments, which connects a user, role, and access scope.

The audit structure separates the main audit event from its field-level changes. Data import belongs to this module because it supports controlled integration rather than representing a standalone manufacturing process.

3.5.3. Product and Production Master Data ERD

This module contains the reusable definitions required before planning or production can begin. It includes units of measure, Items, Bills of Materials, Routings, Routing operations, Work Centers, shifts, work calendars, and standard costs.



Figure 5. ERD of the Product and Production Master Data module.

An Item belongs to a Company and may be used as either a BOM parent or component. A BOM header identifies the manufactured parent Item and revision, while BOM lines define the component Items and their required quantities.

A Routing belongs to an Item and contains ordered Routing operations. Operations may reference Work Centers, while Work Centers may use work calendars composed of weekly shifts and date-specific exceptions. These relationships provide the production structure later snapshotted into Work Orders.

3.5.4. Sales and Planning ERD

The Sales and Planning module connects customer demand with material requirements. It includes Sales Orders, Sales Order lines, planning demands, MRP runs, run-demand snapshots, requirement lines, and supply suggestions.

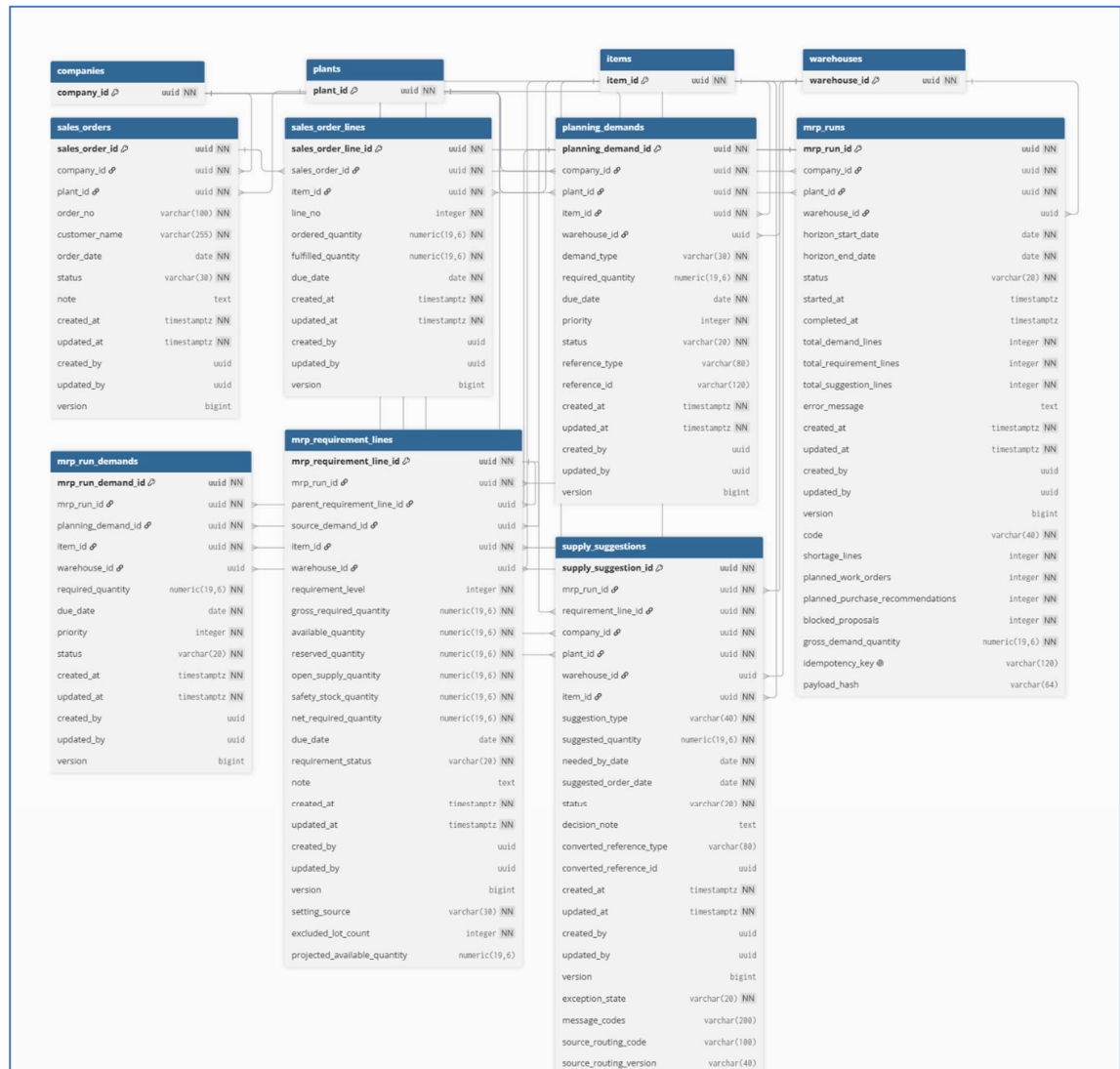


Figure 6. ERD of the Sales and Planning module.

A Sales Order contains one or more Sales Order lines. Confirmed lines generate planning demand, while an MRP run captures an immutable demand snapshot through mrp_run_demands.

Requirement lines may form a hierarchy through parent_requirement_line_id, representing multi-level BOM explosion. A shortage requirement produces a supply suggestion that may recommend either purchasing or internal manufacturing.

3.5.5. Purchasing ERD

The Purchasing module represents external material supply. It includes suppliers, supplier-specific Item data, Purchase Requisitions, Purchase Orders, Goods Receipts, and their corresponding lines.

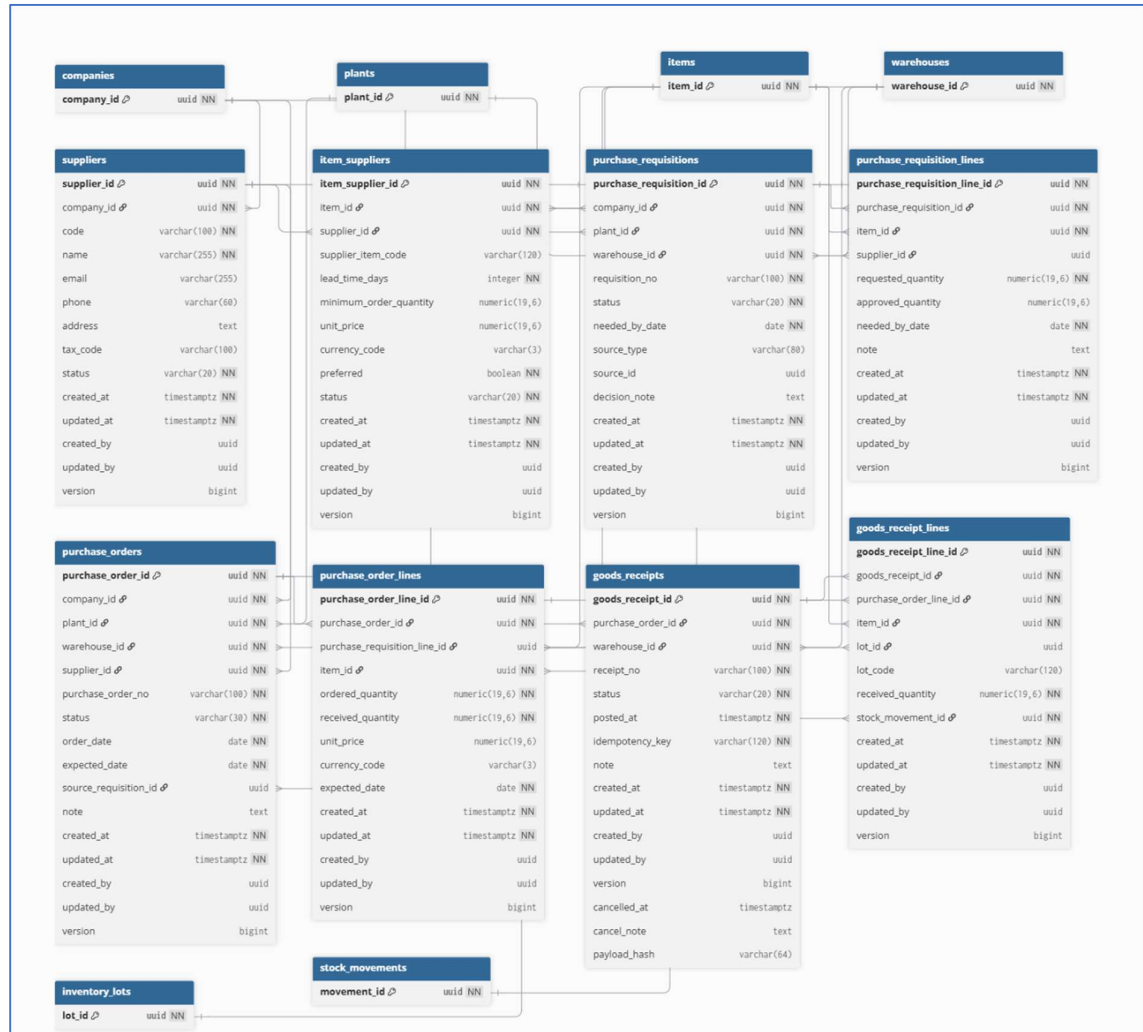


Figure 7. ERD of the Purchasing module.

item_suppliers establishes the sourcing relationship between an Item and Supplier, including lead time, minimum order quantity, price, currency, and preferred-supplier status.

A Purchase Requisition contains requested Item lines and may be converted into a Purchase Order. Purchase Order lines retain their source requisition line where applicable.

Goods Receipt lines reference Purchase Order lines and the inventory movement created by posting the receipt.

3.5.6. Inventory and Warehouse ERD

This module manages item tracking policies, planning settings, inventory balances, lots, serial numbers, and append-only stock movements.



Figure 8. ERD of the Inventory and Warehouse module.

Inventory is identified by Item, Warehouse, and optional lot or serial information. stock_balances provides query-optimized on-hand, reserved, and quality-hold quantities. stock_movements preserves the immutable history of receive, issue, adjustment, reversal, and lot-status transactions.

Item-Warehouse settings store safety stock, reorder point, and lead-time configuration used by planning and operational alerts. Lot and serial statuses determine whether a quantity is eligible for reservation, issue, MRP, or fulfillment.

3.5.7. Manufacturing Execution and Quality ERD

This module records the internal production lifecycle from Work Order creation through material consumption, WIP reporting, finished-goods receipt, and quality disposition.

Figure 9. ERD of the Manufacturing Execution and Quality module.

A Work Order contains immutable component and operation snapshots. Material reservations connect component requirements with eligible Warehouse and lot stock. Material Issues consume those reservations and connect each issue line to its inventory movement.

WIP transactions and Production Executions preserve actual manufacturing evidence. Production Receipt lines connect output Items, Warehouses, lots or serials, and stock movements. Quality dispositions are stored separately so receipt approval and final quality acceptance remain distinct decisions.

3.5.8. Cross-Module Data Relationships

The six modules form one connected production model rather than six isolated databases. The principal cross-module relationships are summarized below.

Relationship chain	Purpose
Company -> Plant -> Warehouse	Establishes organizational ownership and authorization scope.
Item -> BOM / Routing / Inventory / Purchasing	Connects product definition with planning, sourcing, and stock.
Sales Order Line -> Planning Demand -> MRP Run	Preserves the origin of demand entering material planning.
MRP Requirement -> Supply Suggestion	Records shortage analysis and the recommended BUY or MAKE response.
Purchase Order -> Goods Receipt -> Stock Movement	Connects supplier delivery with the inventory ledger.
Work Order -> Material Issue -> Stock Movement	Connects production consumption with inventory.
Work Order -> Production Receipt -> Quality Disposition	Connects manufactured output with approval and QC status.
Work Order Allocation -> Sales Order Line	Connects quality-approved output back to customer demand.

Table 5. Cross-module data relationships.

3.6. Business Rules and State Controls

OmniPlant applies validation at both the document and inventory levels. The principal rules are summarized in the following table.

Rule	Enforced contract
Quantity	Transactional, Sales Order, and production quantities must be greater than zero.
Lot eligibility	HOLD, REJECTED, and expired lots are excluded from availability, reservation, MRP, and fulfillment.
Demand	Only confirmed and open demand participates in planning and allocation.
Snapshot	Work Orders retain immutable BOM and Routing evidence.
Release	All component requirements must be fully reserved before Work Order release.
Issue	Only a posted Material Issue decreases inventory; over-BOM issue requires reason and permission.
Receipt	Submission has no inventory effect; only approval creates finished-goods inventory.
Quality	Approved output remains on HOLD until disposition; only AVAILABLE output satisfies demand.
Mutation safety	Integrity-sensitive commands require confirmation and stable idempotency-key reuse on retry.

Table 6. Core business validation rules.

3.7. API, Security, and Transaction Resilience

Every backend response uses a standard envelope containing a response code, message, and optional result. Paginated endpoints additionally return page, size, total element, and total-page information.

Authentication uses JWT access tokens together with rotating refresh tokens. After authentication, the client loads the authoritative user profile, permissions, role assignments, and access scopes. The frontend uses this information to control navigation and action visibility, while the backend performs the final authorization check for every protected operation.

Authorization combines permissions with organizational scope. A user may have permission to manage Work Orders but remain limited to a specific Plant. Requests outside the assigned scope are rejected even when the general permission is present.

State-changing requests use stable idempotency keys. Retrying the same logical action reuses its original key, preventing duplicate inventory movements, MRP runs, Material Issues, Goods Receipts, or Production Receipts during network failures.

3.8. Frontend Interaction Design

The frontend is designed primarily for desktop and tablet operation. Dense lists, filters, status indicators, and forms follow a consistent ERP interaction model.

Loading states use Skeleton placeholders. Empty states explain why no records are available and display creation actions only when the user has permission. Validation and API errors preserve entered values so the user can correct or retry a request without re-entering the entire form.

Critical state changes require explicit confirmation. These actions include Sales Order confirmation, BOM and Routing activation, Work Order release, Material Issue posting, receipt approval or rejection, and quality disposition.

All user-visible content is maintained in English and Vietnamese translation namespaces. Status labels and validation messages use consistent terminology across list, detail, and transaction screens.

3.9. System Operation and Deployment

A standard local environment consists of PostgreSQL, Redis, the Spring Boot backend, and the Next.js frontend. PostgreSQL migrations are managed through Flyway. The backend normally runs on port 8080, while the frontend runs on port 3000.

The infrastructure must be healthy before operational testing begins. After startup, the tester verifies authentication, role and scope loading, Plant selection, API health, and migration status. End-to-end verification then proceeds through demand, planning, supply conversion, production or purchasing execution, inventory posting, quality disposition, and audit review.

Operational step	Expected check
1. Start infrastructure	PostgreSQL and Redis are healthy and Flyway migrations are complete.
2. Start API	Actuator health reports UP and OpenAPI is reachable.

3. Start frontend	The application loads without console errors and unauthenticated users are redirected to Login.
4. Authenticate	Profile, permissions, scopes, and default Plant are loaded.
5. Select Plant	Queries refetch without displaying stale data from another Plant.
6. Execute workflow	Confirmation and idempotent commands are used and document status is verified after each transition.
7. Verify results	Movements, lot status, Dashboard, Sales Order fulfillment, and audit evidence are checked.

Table 7. Operational startup and verification checklist.

3.10. Chapter Summary

This chapter presented the architectural, functional, process, and data design of OmniPlant. The use case diagram defined the responsibilities of Administrator, Manager, and Operator, while the swimlane activity diagram described the execution sequence and automated System behavior.

The six ERDs demonstrate how administration, master data, sales and planning, purchasing, inventory, and manufacturing execution are separated into readable modules while remaining connected through explicit relationships. Together, these designs establish traceability from organizational access and customer demand to material planning, purchasing or production, inventory posting, and quality-controlled output.

CHAPTER 4. RESULTS AND DISCUSSION

4.1. Implementation Results

Capstone 2 completed the focused ERP production loop from confirmed Sales Order demand to QC-approved AVAILABLE inventory. The implementation is no longer a high-fidelity prototype: the primary MVP routes use backend APIs, UUID identifiers, decimal-safe adapters, permission-aware React Query hooks and documented invalidation across related modules.

Area	Delivered behavior	Status
Planning	Confirmed demand -> immutable run -> requirements/netting -> MAKE/BUY suggestions	Complete and API integrated
Work Order	Conversion/manual draft, plan, snapshots, readiness, reservation, release, cancel/close	Complete and API integrated
Material Issue	Reservation-backed and eligible-stock issue, stock movement, trace and variance seam	Complete and live write accepted
Manual Execution	Good/scrap/rework, actual time, notes and WIP history	Complete and API integrated
Production Receipt	Partial receipt, submit, approve/reject, HOLD and QC disposition	Complete and API integrated
Inventory/Dashb oard	Balances, lots, movements, availability exclusion, alerts and recent activity	Complete and live accepted
Admin/RBAC	Users, roles, permissions, scopes and assignments	Complete for current three-actor model

Table 8. Capstone 2 delivery summary

4.2. API-Backed Interface Evidence

The following subsections present the current OmniPlant interface screenshots embedded in the Capstone 2 presentation. Each subsection documents one screen, its operational purpose and the evidence visible in the refreshed demonstration dataset.

4.2.1. Dashboard Overview

This screen gives managers an API-backed summary of current inventory and manufacturing activity. It highlights operational alerts and recent movements so exceptions can be identified quickly.

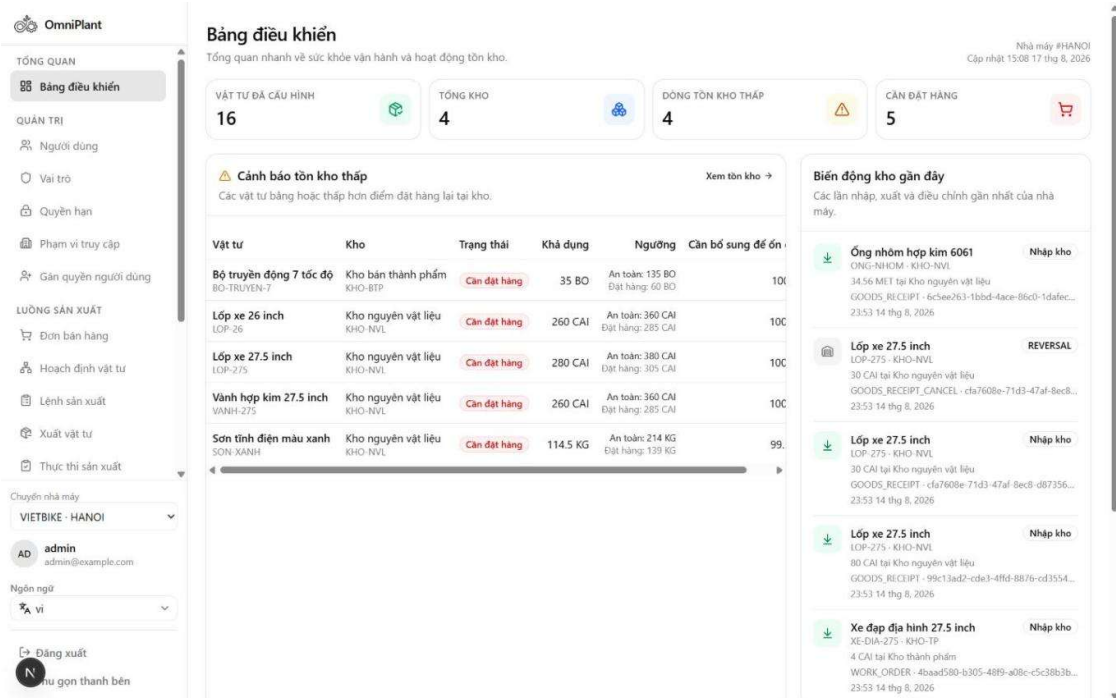


Figure 10. Dashboard with API-backed operational overview and alerts

4.2.2. Item Master

This screen maintains the core item records used across inventory, BOM and production processes. It shows item type, unit of measure and tracking policy to ensure consistent downstream transactions.

Kho	Tồn kho an toàn	Điểm đặt hàng lại	Thời gian chờ (ngày)	Trạng thái
Kho thành phẩm	0.000000	0.000000	3	ACTIVE

Thêm cấu hình kho

Kho:

Tồn kho an toàn: Điểm đặt hàng lại:

Thời gian chờ (ngày):

Figure 11. Item Master with item types, UOM and tracking-policy evidence

4.2.3. Stock Balance

This screen presents inventory quantities by item and storage location. On-hand, reserved and available balances are separated so users can evaluate usable stock accurately.

OmniPlant

TỔNG QUAN

Bảng điều khiển

QUẢN TRỊ

Người dùng

Vai trò

Quyền hạn

Phạm vi truy cập

Gán quyền người dùng

LƯUỠNG SẢN XUẤT

Đơn bán hàng

Hoạch định vật tư

Lệnh sản xuất

Xuất vật tư

Thực thi sản xuất

Chuyển nhà máy

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Ngôn ngữ

vi

Đăng xuất

hu gọn thành bên

Tồn kho & Kho hàng

Dữ liệu tồn kho trực tiếp

Kho

KHO-NVL - Kho nguyên vật liệu

VẬT TƯ	TỒN KHO THỰC	CÓ THỂ DÙNG	ĐÃ ĐẶT TRƯỚC	CHỖ KIỂM ĐỊNH	DANH SÁCH LỖ HÀNG	
ONG-NHOM Ổng nhôm hợp kim 6061	34.56 MET	34.56 MET	0 MET	0 MET	1	Nhập kho Xuất kho Điều chỉnh
LOP-275 Lốp xe 27.5 inch	320 CAI	280 CAI	40 CAI	0 CAI	0	Nhập kho Xuất kho Điều chỉnh
OC-VIT-M6 Ốc vít M6 mạ kẽm	5000 CAI	4608.32 CAI	391.68 CAI	0 CAI	0	Nhập kho Xuất kho Điều chỉnh
PHANH-V Bộ phanh V-brake	200 BO	150 BO	50 BO	0 BO	0	Nhập kho Xuất kho Điều chỉnh
TAY-LAI Tay lái nhôm	200 CAI	175 CAI	25 CAI	0 CAI	0	Nhập kho Xuất kho Điều chỉnh
YEN-XE Yên xe đa tổng hợp	200 CAI	171 CAI	29 CAI	0 CAI	0	Nhập kho Xuất kho Điều chỉnh
DONG-CO Động cơ trợ lực 250W	20 CAI	16 CAI	4 CAI	0 CAI	0	Nhập kho Xuất kho Điều chỉnh
PIN-36V Pin Li-ion 36V	20 CAI	16 CAI	4 CAI	0 CAI	1	Nhập kho Xuất kho Điều chỉnh
LOP-26 Lốp xe 26 inch	260 CAI	260 CAI	0 CAI	0 CAI	0	Nhập kho Xuất kho Điều chỉnh
MOAYO Moay-ơ vòng bi kín	400 CAI	360 CAI	40 CAI	0 CAI	0	Nhập kho Xuất kho Điều chỉnh
NAN-HOA Nan hoa thép không gỉ	9000 CAI	7656 CAI	1344 CAI	0 CAI	0	Nhập kho Xuất kho Điều chỉnh

Figure 12. Stock Balance with on-hand, reserved and available quantities

4.2.4. Inventory Lots

This screen provides lot-level inventory visibility for traceable materials and products. Status values such as AVAILABLE, HOLD and REJECTED make quality eligibility clear before planning or issue.

OmniPlant

TỔNG QUAN

- Bảng điều khiển

QUẢN TRỊ

- Người dùng
- Vai trò
- Quyền hạn
- Phạm vi truy cập
- Gán quyền người dùng

LƯU TRÌNH SẢN XUẤT

- Đơn bán hàng
- Hoạch định vật tư
- Lệnh sản xuất
- Xuất vật tư
- Thực thi sản xuất

Chuyển nhà máy

- VIETBIKE - HANOI

adminadmin@example.com

Ngôn ngữ

- vi

Đăng xuất

Nhau gọn thành bên

Lô tồn kho

Theo dõi lô, hạn dùng và trạng thái khả dụng.

Q. Tìm lô, SKU hoặc vật tư...

KHO-BTP - Kho bán thành phẩm

Tất cả trạng thái

Đặt lại

Vật tư	Lô	Tồn thực tế	Khả dụng	Trạng thái	Hạn dùng	
Bánh xe 26 inch hoàn chỉnh BARH-26	LO-BANH26-01	70	38	Khả dụng	—	...
Khung nhôm xe phở (đã sơn) KHUNG-PHO	LO-KHUNGPHO-01	30	24	Khả dụng	—	...
Khung nhôm xe địa hình (đã sơn) KHUNG-DIA	LO-KHUNGDIA-01	36	27	Khả dụng	—	...
Bánh xe 27.5 inch hoàn chỉnh BARH-275	LO-BANH275-01	40	14	Khả dụng	—	...
Bánh xe 27.5 inch hoàn chỉnh BARH-275	LO-BANH275-LOI	1	0	Từ chối	—	...
Bánh xe 27.5 inch hoàn chỉnh BARH-275	LO-BANH275-DAT	1	1	Khả dụng	—	...
Bánh xe 27.5 inch hoàn chỉnh BARH-275	LO-BANH275-GIU	1	0	Tạm giữ	—	...
Ống thép Crôm-Môlipden 4130 ONG-THIEP	LO-THIEP-LOI	40	0	Từ chối	—	...

Số dòng mỗi trang20

Hiện thị 1 đến 8 trong tổng số 8 kết quả

Quay lại

Tiếp

Figure 13. Inventory Lots with AVAILABLE, HOLD and REJECTED status evidence

4.2.5. Stock Movement History

This screen records the append-only history of inventory transactions. Movement types and trace references help users audit how stock quantities changed over time.

OmniPlant

TỔNG QUAN

- Bảng điều khiển

QUẢN TRỊ

- Người dùng
- Vai trò
- Quyền hạn
- Phạm vi truy cập
- Gán quyền người dùng

LƯUỒNG SẢN XUẤT

- Đơn bán hàng
- Hoạch định vật tư
- Lệnh sản xuất
- Xuất vật tư
- Thực thi sản xuất

Chuyển nhà máy

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- vi

Đăng xuất

Như gọn thành bên

←Biến động kho

Xem mọi thay đổi tồn kho và chi tiết truy vết:

KHO-NVL - Kho nguyên vật liệu

TIME	TYPE	ITEM	WAREHOUSE	LOT	QUANTITY	TRACE
8/14/2026, 11:53:25 PM	RECEIVE	ONG-NHOM - Ống nhôm hợp kim 6061	KHO-NVL	LO-NHOM-03	34.56 MET	DEPO-NHAPPHUA-3004-1:1:1
8/14/2026, 11:53:24 PM	REVERSAL	LOP-275 - Lốp xe 27.5 inch	KHO-NVL	—	30 CAI	cfa76086-71d3-472f-bac8-d873568baffe::cancel:1:1
8/14/2026, 11:53:24 PM	RECEIVE	LOP-275 - Lốp xe 27.5 inch	KHO-NVL	—	30 CAI	DEPO-NHAPPHUA-3003-2:1:1
8/14/2026, 11:53:24 PM	RECEIVE	LOP-275 - Lốp xe 27.5 inch	KHO-NVL	—	80 CAI	DEPO-NHAPPHUA-3003-1:1:1
8/14/2026, 11:52:54 PM	ISSUE	VANH-275 - Vành hợp kim 27.5 inch	KHO-NVL	—	20 CAI	DEPO-XUAT-MO-2605-1:1:1
8/14/2026, 11:52:49 PM	LOT_STATUS_CHANGE	SON-XANH - Sơn tĩnh điện màu xanh	KHO-NVL	LO-SON-GIU	60 KG	DEPO-TRANGTHA1LO-LO-SON-GIU
8/14/2026, 11:52:47 PM	RECEIVE	OC-VIT-M6 - Ốc vít M6 mạ kẽm	KHO-NVL	—	5000 CAI	DEPO-NHAP-HANOI-OC-VIT-M6-22
8/14/2026, 11:52:47 PM	RECEIVE	KHI-HAN - Khí hàn Argon	KHO-NVL	—	400 LIT	DEPO-NHAP-HANOI-KHI-HAN-21
8/14/2026, 11:52:47 PM	RECEIVE	SON-XANH - Sơn tĩnh điện màu xanh	KHO-NVL	LO-SON-GIU	60 KG	DEPO-NHAP-HANOI-SON-XANH-20
8/14/2026, 11:52:47 PM	RECEIVE	SON-XANH - Sơn tĩnh điện màu xanh	KHO-NVL	LO-SON-01	120 KG	DEPO-NHAP-HANOI-SON-XANH-19
8/14/2026, 11:52:47 PM	RECEIVE	DONG-CO - Động cơ trợ lực 250W	KHO-NVL	—	20 CAI	DEPO-NHAP-HANOI-DONG-CO-18

Figure 14. Stock Movement history with transaction types and trace references

4.2.6. BOM List

This screen lists Bills of Materials and their lifecycle states. Users can distinguish active production definitions from drafts before selecting a structure for planning or execution.

OmniPlant

Công thức sản xuất (BOM)
Quản lý định nghĩa BOM, phiên bản và trạng thái sản phẩm.

+ Tạo BOM

Q Tìm theo mã BOM, tên hoặc SKU sản phẩm ALL Đặt lại

Sản phẩm đầu ra	Bản sửa đổi	Thành phần	Trạng thái	Cập nhật lần cuối
BANH-275 Bánh xe 27.5 inch hoàn chỉnh	A	4	ACTIVE	8/14/2026
BANH-26 Bánh xe 26 inch hoàn chỉnh	A	4	ACTIVE	8/14/2026
KHUNG-DIA Khung nhôm xe địa hình (đã sơn)	A	3	ACTIVE	8/14/2026
KHUNG-PHO Khung nhôm xe phở (đã sơn)	A	3	ACTIVE	8/14/2026
XE-DIEN Xe đạp điện trợ lực 36V	A	5	ACTIVE	8/14/2026
XE-TRE-EM Xe đạp trẻ em 16 inch	A	4	ACTIVE	8/14/2026
XE-DIA-275 Xe đạp địa hình 27.5 inch	A	6	ACTIVE	8/14/2026
XE-PHO-26 Xe đạp phở 26 inch	B	7	DRAFT	8/14/2026
XE-PHO-26 Xe đạp phở 26 inch	A	7	ACTIVE	8/14/2026

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Đăng xuất
hu gọn thành bên

Figure 15. BOM list with active and draft lifecycle visibility

4.2.7. Active BOM Detail

This screen displays the approved product structure for a manufactured item. Revision, output quantity and component lines define the material basis used by planning and Work Orders.

OmniPlant ← Công thức sản xuất (BOM)

ACTIVE A Vô hiệu hóa

Xe đạp phở 26 inch

XE-PHO-26

Bản sửa đổi
A

Thành phần
7

Cập nhật lần cuối
8/14/2026

Mô tả
Định mức sản xuất XE-PHO-26 bản A

Thành phần

Dòng	Vật tư thành phần	Số lượng trên một đơn vị	Tỷ lệ hao hụt
1	KHUNG-PHỞ Khung nhôm xe phở (đã sơn)	1	0%
2	BÁNH-26 Bánh xe 26 inch hoàn chỉnh	2	0%
3	BỘ-TRUYỀN-7 Bộ truyền động 7 tốc độ	1	0%
4	YÊN-XE Yên xe đa tổng hợp	1	0%
5	TAY-LÁI Tay lái nhôm	1	0%
6	PHANH-V Bộ phanh V-brake	2	0%
7	ỐC-VÍT-M6 Ốc vít M6 mạ kẽm	24	2%

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vi

Đăng xuất
Nhấn vào đây để đăng xuất

Figure 16. Active BOM detail with revision, output quantity and component lines

4.2.8. Multi-Level BOM Tree

This screen visualizes nested assemblies and components across multiple BOM levels. The hierarchy helps users understand dependent quantities and the full material composition of the finished product.

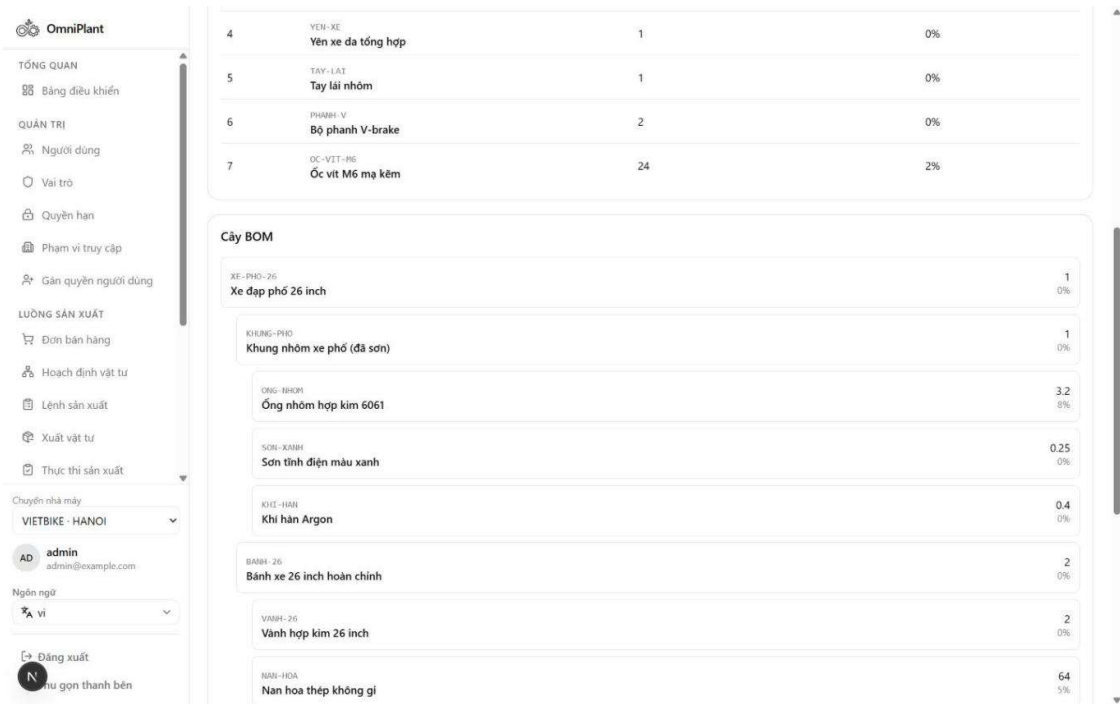


Figure 17. Multi-level BOM tree showing nested sub-assemblies and component quantities

4.2.9. Sales Order List

This screen summarizes customer demand and its document lifecycle. Order and fulfillment statuses allow managers to identify confirmed demand that is ready for material planning.

Đơn bán hàng
Theo dõi nhu cầu khách hàng, phân bổ sản xuất và tiến độ đáp ứng.

Tất cả trạng thái

ĐƠN BÁN HÀNG	KHÁCH HÀNG / THAM CHIẾU	NGÀY ĐẶT HÀNG	ĐỒNG NHU CẦU	TRANG THÁI	
SO-1003	Đại lý Xe đạp Hoàng Anh	2026-08-14	1	CANCELLED	
SO-1002	Siêu thị Thể thao Đồng Lực	2026-08-14	1	CONFIRMED	Hủy đơn hàng
SO-1007	Trường Tiểu học Kim Liên	2026-08-14	1	CONFIRMED	Hủy đơn hàng
SO-1001	Cửa hàng Xe đạp Thống Nhất	2026-08-14	1	DRAFT	Sửa bản nháp Xác nhận đơn Hủy đơn hàng
SO-1005	Xe đạp Sài Gòn	2026-08-14	1	FULFILLED	
SO-1006	Siêu thị Thể thao Đồng Lực	2026-08-14	1	IN PRODUCTION	
SO-1004	Cửa hàng Xe đạp Thống Nhất	2026-08-14	1	PARTIALLY_FULFILLED	

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Đăng xuất
Thu gọn thanh bên

Figure 18. Sales Order list with lifecycle and fulfillment status evidence

4.2.10. Sales Order Detail

This screen shows the selected order, its line items and requested quantities. Allocation and fulfillment values connect customer demand with later planning and production results.

The screenshot displays the 'Sales Order Detail' screen in the OmniPlant system. The sidebar on the left contains navigation links for 'TỔNG QUAN', 'QUẢN TRỊ', and 'LƯUỒNG SẢN XUẤT'. The main content area shows the order status as 'PARTIALLY FULFILLED' and 'SO-1004'. The order is for 'Cửa hàng Xe đạp Thống Nhất' and was created on '2026-08-14'. A table lists the line items, showing one item: 'XE-PHO-26 - Xe đạp phở 26 inch' with a quantity of 10, allocated 6, and a delivery date of 2026-09-18.

DÒNG	VẬT TƯ	ĐÃ ĐẶT	ĐÃ PHÂN BỐ	ĐÃ HOÀN THÀNH	HẠN GIAO
1	XE-PHO-26 - Xe đạp phở 26 inch	10 CẢI	6	6	2026-09-18

Figure 19. Sales Order detail with line allocation and fulfillment quantities

4.2.11. Material Planning Workspace

This screen is the entry point for converting confirmed demand into an MRP Planning Run. It presents eligible Sales Orders and previous run history for controlled, auditable planning.

OmniPlant

TỔNG QUAN

Bảng điều khiển

QUẢN TRỊ

Người dùng

Vai trò

Quyền hạn

Phạm vi truy cập

Gán quyền người dùng

LƯUỒNG SẢN XUẤT

Đơn bán hàng

Hoạch định vật tư

Lệnh sản xuất

Xuất vật tư

Thực thi sản xuất

Chuyển nhà máy

VIETBIKE - HANOI

admin
admin@example.com

Nguồn ngữ

vi

Đăng xuất

Như gọn thành bên

Hoạch định vật tư

Chuyển nhu cầu đã xác nhận thành kế hoạch vật tư đa cấp bất biến với đề xuất MAKE và BUY.

Bắt đầu kỳ

08/17/2026

Kết thúc kỳ

10/16/2026

Kho hoạch định

KHO-TP - Kho thành phẩm

Nhu cầu đã xác nhận

Chọn các dòng Đơn bán hàng còn mở và đến hạn trong kỳ hoạch định.

ĐƠN BÁN HÀNG

THÀNH PHẨM

SỐ LƯỢNG CÒN MỞ

NGÀY CẦN

☐

SO-1004 - 1

XE-PHO-26
Xe đạp phở 26 inch

4 CAI

2026-09-18

☐

SO-1002 - 1

XE-DIA-275
Xe đạp địa hình 27.5 inch

8 CAI

2026-09-28

☐

SO-1006 - 1

XE-PHO-26
Xe đạp phở 26 inch

6 CAI

2026-10-03

☐

SO-1007 - 1

XE-TRE-EM
Xe đạp trẻ em 16 inch

12 CAI

2026-10-13

Đã chọn 0 dòng nhu cầu

Chạy hoạch định

Lịch sử Planning Run

RUN-79966D84

2026-08-14 -- 2026-11-12

9 Planned WO - 11 Planned PO

RUN-88082621

2026-08-14 -- 2026-11-12

13 Planned WO - 36 Planned PO

Figure 20. Material Planning workspace with confirmed demand and Planning Run history

4.2.12. Planning Run Detail

This screen explains the calculated material requirements for a specific immutable Planning Run. Shortages and MAKE or BUY suggestions show how available supply is compared with production demand.

The screenshot displays the OmniPlant Planning Run Detail interface. On the left is a sidebar with navigation options: TÓNG QUAN, Bảng điều khiển, QUẢN TRỊ, Người dùng, Vai trò, Quyền hạn, Phạm vi truy cập, Gán quyền người dùng, LƯUỒNG SẢN XUẤT, Đơn bán hàng, and Hoạch định vật tư (highlighted). Below the sidebar are filters for Chuyển nhà máy (VIETBIKE - HANOI), user (admin), and ngôn ngữ (vi). The main area shows a list of items with their details and supply decisions:

- BANH-26 · Bánh xe 26 inch hoàn chỉnh** (20 BO - 2026-09-15): Cần xem xét. Notes: • Tồn AVAILABLE không đáp ứng nhu cầu rõ ràng. • Một hoặc nhiều tham số đang đúng cấu hình mặc định hệ thống. Buttons: Nhập, Từ chối, Duyệt.
- BO-TRUYEN-7 · Bộ truyền động 7 tốc độ** (10 BO - 2026-09-15): Bị chặn. Notes: • Tồn AVAILABLE không đáp ứng nhu cầu rõ ràng. • Item MAKE chưa có BOM ACTIVE. • Chưa có Routing ACTIVE trong Nhà máy. • Một hoặc nhiều tham số đang đúng cấu hình mặc định hệ thống. Buttons: Nhập, Từ chối, Duyệt.
- XE-PHO-26 · Xe đạp phở 26 inch** (10 CAI - 2026-09-18): Sẵn sàng. Notes: • Tồn AVAILABLE không đáp ứng nhu cầu rõ ràng. Buttons: Đã chuyển đổi, Xem Work Order.
- XE-DIA-275 · Xe đạp địa hình 27.5 inch** (4 CAI - 2026-09-23): Cần xem xét. Notes: • Tồn AVAILABLE không đáp ứng nhu cầu rõ ràng. • Một hoặc nhiều tham số đang đúng cấu hình mặc định hệ thống. Buttons: Đã chuyển đổi, Xem Work Order.
- KHUNG-DIA · Khung nhôm xe địa hình (đã sơn)** (4 CAI - 2026-09-23): Cần xem xét. Notes: • Tồn AVAILABLE không đáp ứng nhu cầu rõ ràng. • Một hoặc nhiều tham số đang đúng cấu hình mặc định hệ thống. Buttons: Nhập, Từ chối, Duyệt.
- KHI-HAN · Khí hàn Argon** (4 LIT - 2026-09-15): Cần xem xét. Notes: • Tồn AVAILABLE không đáp ứng nhu cầu rõ ràng. • Một hoặc nhiều tham số đang đúng cấu hình mặc định hệ thống. Buttons: Nhập, Từ chối, Duyệt.
- VANH-26 · Vành hợp kim 26 inch** (40 CAI - 2026-09-15): Cần xem xét. Notes: • Tồn AVAILABLE không đáp ứng nhu cầu rõ ràng. • Một hoặc nhiều tham số đang đúng cấu hình mặc định hệ thống. Buttons: Nhập, Từ chối, Duyệt.
- NAN-HOA · Nan hoa thép không gỉ** (1344 CAI - 2026-09-15): Cần xem xét. Notes: • Tồn AVAILABLE không đáp ứng nhu cầu rõ ràng. • Một hoặc nhiều tham số đang đúng cấu hình mặc định hệ thống. Buttons: Đã duyệt.
- MOAYO · Moay-ơ vòng bi kín** (40 CAI - 2026-09-15): Cần xem xét. Notes: • Tồn AVAILABLE không đáp ứng nhu cầu rõ ràng. • Một hoặc nhiều tham số đang đúng cấu hình mặc định hệ thống. Buttons: Nhập, Từ chối, Duyệt.
- LOP-26 · Lốp xe 26 inch** (40 CAI - 2026-09-15): Cần xem xét. Notes: • Tồn AVAILABLE không đáp ứng nhu cầu rõ ràng. • Một hoặc nhiều tham số đang đúng cấu hình mặc định hệ thống. Buttons: Nhập, Từ chối, Duyệt.

Figure 21. Planning Run detail with shortage and MAKE/BUY supply decisions

4.2.13. Work Order List

This screen provides an overview of manufacturing orders and their current lifecycle states. Users can monitor which orders are planned, released, in progress, blocked or completed.

OmniPlant

TỔNG QUAN

Bảng điều khiển

QUẢN TRỊ

Người dùng

Vai trò

Quyền hạn

Phạm vi truy cập

Gán quyền người dùng

LƯUỒNG SẢN XUẤT

Đơn bán hàng

Hoạch định vật tư

Lệnh sản xuất

Xuất vật tư

Thực thi sản xuất

Chuyển nhà máy

VIETBIKE - HANOI

AD admin admin@example.com

Ngôn ngữ

vi

Đăng xuất

Như gọn thành bên

Work Order

Lệnh sản xuất kế hoạch được tạo từ hoạch định vật tư.

13 Work Order

Tạo thủ công

Tim Work Order, Mặt hàng hoặc Planning Run

Tất cả trạng thái

Tất cả thành phẩm

WORK ORDER	ĐẦU RA	SỐ LƯỢNG	LỊCH	SNAPSHOT	TRẠNG THÁI	
WO-2612 RUN-88082621	XE-PHO-26 Xe đạp phở 26 inch	6 CAI	2026-08-17T06:00:00.000Z 2026-08-22T22:00:00.000Z	BOM revision A QT-XE-PHO v1	RELEASED	Xem
WO-2611 RUN-88082621	XE-DIA-275 Xe đạp địa hình 27.5 inch	4 CAI	2026-08-17T06:00:00.000Z 2026-08-22T22:00:00.000Z	BOM revision A QT-XE-DIA v1	COMPLETED	Xem
WO-2610 RUN-88082621	XE-PHO-26 Xe đạp phở 26 inch	10 CAI	2026-08-17T06:00:00.000Z 2026-08-22T22:00:00.000Z	BOM revision A QT-XE-PHO v1	IN_PROGRESS	Xem
WO-2620 ---	XE-DIEN Xe đạp điện trợ lực 36V	4 CAI	2026-08-15T06:00:00.000Z ---	BOM revision A QT-XE-DIEN v1	IN_PROGRESS	Xem
WO-2608 ---	KHUNG-PHO Khung nhôm xe phở (đã sơn)	4 CAI	---	BOM revision A QT-KHUNG-PHO v1	CANCELLED	Xem
WO-2607 ---	BANH-26 Bánh xe 26 inch hoàn chỉnh	6 BO	2026-08-15T06:00:00.000Z ---	BOM revision A QT-BANH-26 v1	CLOSED	Xem
WO-2606 ---	XE-DIA-275 Xe đạp địa hình 27.5 inch	5 CAI	2026-08-15T06:00:00.000Z ---	BOM revision A QT-XE-DIA v1	COMPLETED	Xem
WO-2605 ---	BANH-275 Bánh xe 27.5 inch hoàn chỉnh	20 BO	2026-08-15T06:00:00.000Z ---	BOM revision A QT-BANH-275 v1	IN_PROGRESS	Xem
WO-2609 ---	KHUNG-PHO Khung nhôm xe phở (đã sơn)	10 CAI	2026-08-17T06:00:00.000Z 2026-08-19T22:00:00.000Z	BOM revision A QT-KHUNG-PHO v1	RELEASED	Xem
WO-2604 ---	KHUNG-DIA Khung nhôm xe địa hình (đã sơn)	10 CAI	2026-08-17T06:00:00.000Z 2026-08-19T22:00:00.000Z	BOM revision A QT-KHUNG-DIA v1	RELEASED	Xem
WO-2603 ---	XE-TRE-EM Xe đạp trẻ em 16 inch	8 CAI	---	BOM revision A Ghi nhận tổng hợp - Không có Routing snapshot	BLOCKED	Xem

Figure 22. Work Order list with production lifecycle status evidence

4.2.14. Work Order Detail

This screen consolidates the production scope, snapshots and material readiness of a Work Order. Reservation status supports the decision to release the order safely for execution.

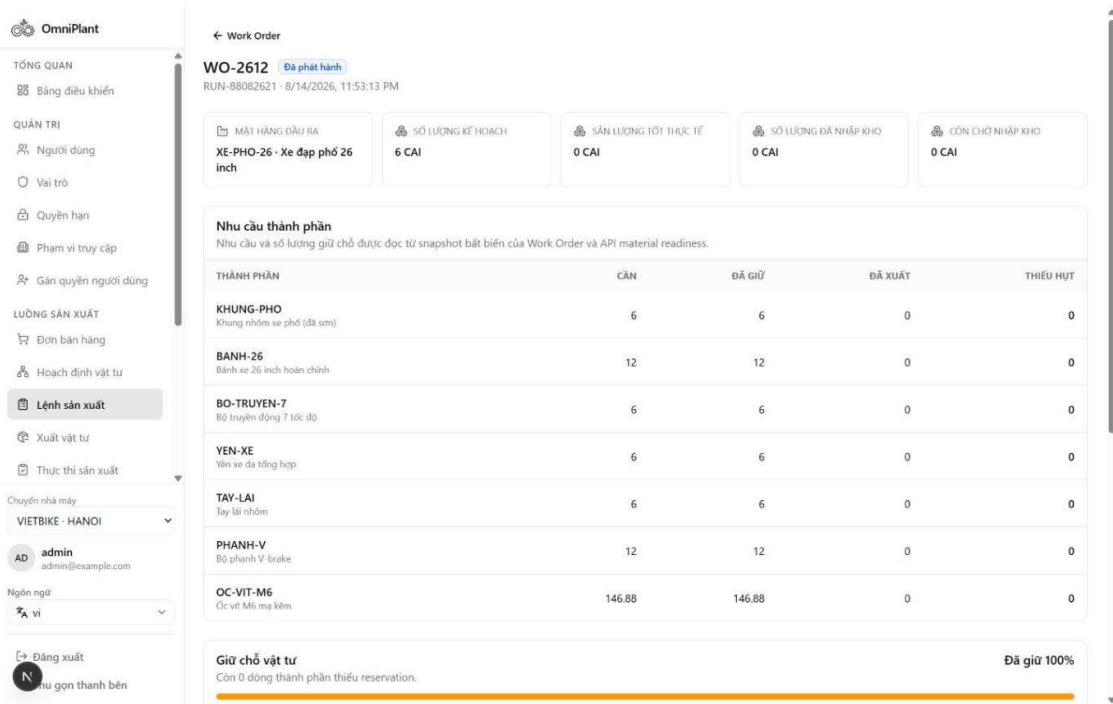


Figure 23. Work Order detail with complete reservation readiness

4.2.15. Material Issue Workspace

This screen supports controlled consumption of reserved components for a Work Order. Component requirements and issue history help operators verify eligible quantities and preserve inventory traceability.

OmniPlant

TỔNG QUAN

Bảng điều khiển

QUẢN TRỊ

Người dùng

Vai trò

Quyền hạn

Phạm vi truy cập

Gán quyền người dùng

LƯUỒNG SẢN XUẤT

Đơn bán hàng

Hoạch định vật tư

Lệnh sản xuất

Xuất vật tư

Thực thi sản xuất

Chuyển nhà máy

VIETBIKE - HANOI

admin
admin@example.com

Ngôn ngữ

vi

Đăng xuất

Thư gọn thành bên

Xuất vật tư

Xuất vật tư đã giữ cho lệnh sản xuất, có truy vết kho, lô, tồn kho và WIP.

Ghi nhận xuất vật tư

Chỉ reservation còn mở của lệnh RELEASED hoặc IN_PROGRESS được phép xuất.

Lệnh sản xuất

WO-2610 - XE-PHO-26 - IN_PROGRESS

Ghi chú phiếu

Ghi chú không bắt buộc cho phiếu xuất vật tư

VẬT TƯ	KHO / LÔ	RESERVATION CÒN LẠI	ĐỊNH MỨC BOM CÒN LẠI	SỐ LƯỢNG XUẤT	LÝ DO	LÝ DO VƯỢT ĐỊNH MỨC
OC-VIT-M6 Ốc vít M6 mạ kẽm	KHO-NVL - Không theo dõi lô	244.8 CAI	244.8 CAI	0	Ghi chú hoặc lý do xuất ()	—
PHANH-V Bộ phanh V-brake	KHO-NVL - Không theo dõi lô	20 BO	20 BO	0	Ghi chú hoặc lý do xuất ()	—
TAY-LAI Tay lái nhôm	KHO-NVL - Không theo dõi lô	10 CAI	10 CAI	0	Ghi chú hoặc lý do xuất ()	—
YEN-XE Yên xe đa tổng hợp	KHO-NVL - Không theo dõi lô	10 CAI	10 CAI	0	Ghi chú hoặc lý do xuất ()	—
BO-TRUYEN-7 Bộ truyền động 7 tốc độ	KHO-BTP - Không theo dõi lô	10 BO	10 BO	0	Ghi chú hoặc lý do xuất ()	—
BANH-26 Bánh xe 26 inch hoàn chỉnh	KHO-BTP - LO-BANH26-01	20 BO	20 BO	0	Ghi chú hoặc lý do xuất ()	—

⚠️ Xuất vượt định mức không dùng reservation

Chỉ dành cho Manager: xuất tồn AVAILABLE hợp lệ mà không tiêu thụ reservation. Số lượng phải làm tổng xuất vượt phần định mức BOM còn lại.

Vật tư / Kho / Lô	Tồn AVAILABLE	Số lượng xuất	Lý do vượt định mức
Chọn vị trí tồn kho hợp lệ	0	0	Bắt buộc khi xuất vượt định mức BOM

Figure 24. Material Issue workspace with component and issue-history evidence

4.2.16. Production Execution Workspace

This screen records manual shop-floor results without claiming MES integration. Operators capture good, scrap and rework quantities together with actual execution evidence.

OmniPlant

TỔNG QUAN

- Bảng điều khiển

QUẢN TRỊ

- Người dùng
- Vai trò
- Quyền hạn
- Phạm vi truy cập
- Gán quyền người dùng

LƯỜNG SẢN XUẤT

- Đơn bán hàng
- Hoạch định vật tư
- Lệnh sản xuất
- Xuất vật tư

Thực thi sản xuất

Chuyển nhà máy

- VIETBIKE - HANOI

adminadmin@example.com

Ngôn ngữ

- vi

Đăng xuất

hu gọn thanh bên

Ghi nhận sản xuất

Ghi nhận thủ công kết quả lệnh sản xuất và WIP, không tích hợp MES.

Ghi nhận kết quả sản xuất

Nhập good, scrap hoặc rework đã xác nhận cho lệnh sản xuất IN_PROGRESS.

Lệnh sản xuất

WO-2605 - BANH-275 - 10/20 BO

Số lượng đạt

Số lượng phế phẩm

Số lượng làm lại

Bắt đầu thực tế

08/17/2026 03:13 PM

Kết thúc thực tế

08/17/2026 03:13 PM

Công đoạn Routing (không bắt buộc)

Kết quả tổng hợp Work Order

Ghi chú

Ghi chú sản xuất, phế phẩm hoặc làm lại (không bắt buộc)

Sản lượng kế hoạch	Lũy kế đạt	Lũy kế phế phẩm	Sản lượng đạt còn lại
20 BO	10 BO	1 BO	10 BO
Cả thể nhập kho	Đã nhập kho	Tương thái Work Order	Tiến độ sản lượng đạt
7 BO	3 BO	IN_PROGRESS	50%

Routing snapshot - chỉ đọc

10. Đan nan hoa20. Cẩn vành và căng nan

Ghi nhận kết quả

Lịch sử kết quả WIP

Các lần xác nhận thủ công kèm operator và tham chiếu truy vết.

BẢN GHI WIP	CÔNG ĐOẠN	ĐẠT	PHẾ PHẨM	LÀM LẠI	OPERATOR	THỜI GIAN THỰC TẾ	TRACE ID
-------------	-----------	-----	----------	---------	----------	-------------------	----------

Figure 25. Production Execution workspace with good, scrap and rework quantities

4.2.17. Production Receipt List

This screen tracks finished-output receipt documents and their approval states. It supports the transition from reported production into HOLD inventory pending quality disposition.

OmniPlant

TỔNG QUAN

Bảng điều khiển

QUẢN TRỊ

Người dùng

Vai trò

Quyền hạn

Phạm vi truy cập

Gán quyền người dùng

LƯỞNG SẢN XUẤT

Đơn bán hàng

Hoạch định vật tư

Lệnh sản xuất

Xuất vật tư

Thực thi sản xuất

Chuyển nhà máy

VIETBIKE - HANOI

admin

admin@example.com

Ngôn ngữ

vi

Đăng xuất

hu gọn thành bên

Nhập kho thành phẩm

Nhận sản lượng đã xác nhận qua phê duyệt, tồn HOLD và QC disposition.

Tạo phiếu nhập từng phần

Tạo phiếu DRAFT từ good WIP đã xác nhận. Submit và phê duyệt là hai thao tác riêng.

Lệnh sản xuất

Chọn lệnh có sản lượng đạt

Kho đích

Chọn kho

Số lượng nhập

Lô đầu ra

Không bắt buộc với item không theo dõi lô

Ghi chú

Tạo phiếu DRAFT

Hàng đợi phiếu nhập, phê duyệt và QC

Operator tạo và submit; Manager phê duyệt/từ chối và thực hiện QC disposition.

PHIẾU NHẬP	LỆNH SẢN XUẤT	SỐ LƯỢNG	KHO	TRANG THÁI PHIẾU	LÔ ĐẦU RA / QC	TRACE ID	TRACE WIP NGUỒN	THAO TÁC	
PR-4E2343C7	WO-2611	4 CẢI	KHO-TP	APPROVED	Đầu ra không theo dõi lô	AVAILABLE	e188e25cf38742d4	ae747bc2b4c342c5	
PR-1A94350A	WO-2610	6 CẢI	KHO-TP	APPROVED	LO-XEPHO-2610	AVAILABLE	67762b4ad1724b9b	d1b09b21f6464ea4	
PR-4F23D72A	WO-2620	1 CẢI	KHO-TP	APPROVED	Đầu ra không theo dõi lô	REJECTED	6ff0b8d58e774b64	a1b954a5ef28b4857	

Figure 26. Production Receipt list with approval and QC lifecycle states

4.2.18. Inventory Lots after Quality Control

This screen shows the quality outcome of received production lots. AVAILABLE, HOLD and REJECTED statuses demonstrate which output can be used for planning or fulfillment.

OmniPlant

TỔNG QUAN

Bảng điều khiển

QUẢN TRỊ

Người dùng

Vai trò

Quyền hạn

Phạm vi truy cập

Gán quyền người dùng

LƯỜNG SẢN XUẤT

Đơn bán hàng

Hoạch định vật tư

Lệnh sản xuất

Xuất vật tư

Thực thi sản xuất

Chuyển nhà máy

VIETBIKE - HANOI

admin

admin@example.com

Ngôn ngữ

vi

Đăng xuất

hu gọn thành bên

Lô tồn kho

Theo dõi lô, hạn dùng và trạng thái khả dụng.

Q. Tìm lô, SKU hoặc vật tư...

KHO-BTP - Kho bán thành phẩm

Tất cả trạng thái

Đặt lại

Vật tư	Lô	Tồn thực tế	Khả dụng	Trạng thái	Hạn dùng
Bánh xe 26 inch hoàn chỉnh BANH-26	LO-BANH26-01	70	38	Khả dụng	—
Khung nhôm xe phở (đã sơn) KHUNG-PHO	LO-KHUNGPHO-01	30	24	Khả dụng	—
Khung nhôm xe địa hình (đã sơn) KHUNG-DIA	LO-KHUNGDIA-01	36	27	Khả dụng	—
Bánh xe 27.5 inch hoàn chỉnh BANH-275	LO-BANH275-01	40	14	Khả dụng	—
Bánh xe 27.5 inch hoàn chỉnh BANH-275	LO-BANH275-LOI	1	0	Từ chối	—
Bánh xe 27.5 inch hoàn chỉnh BANH-275	LO-BANH275-DAT	1	1	Khả dụng	—
Bánh xe 27.5 inch hoàn chỉnh BANH-275	LO-BANH275-GIU	1	0	Tạm giữ	—
Ống thép Crôm-Môlipden 4130 ONG-THIEP	LO-THIEP-LOI	40	0	Từ chối	—

Số dòng mỗi trang

20

Hiện thị 1 đến 8 trong tổng số 8 kết quả

Quay lại

Tiếp

Figure 27. Inventory Lots after QC with AVAILABLE, HOLD and REJECTED output

4.2.19. Login and Session Initialization

This screen authenticates the user and initializes the authorized operating context. Session and Plant scope are established before protected ERP functions become available.

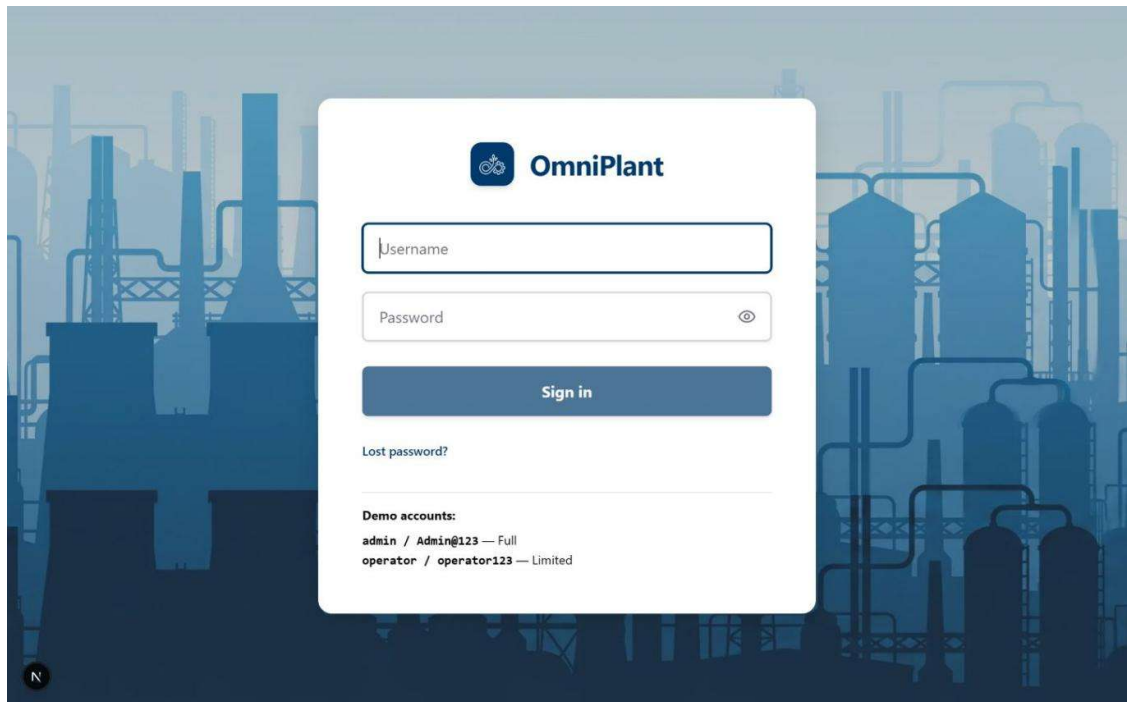


Figure 28. Login interface for session and Plant-scope initialization

4.2.20. User Management

This screen maintains application accounts and their lifecycle status. Administrators can review active or inactive users before assigning responsibilities and access scope.

OmniPlant

TỔNG QUAN

- Bảng điều khiển

QUẢN TRỊ

- Người dùng**
- Vai trò
- Quyền hạn
- Phạm vi truy cập
- Gán quyền người dùng

LƯUỒN SẢN XUẤT

- Đơn bán hàng
- Hoạch định vật tư
- Lệnh sản xuất
- Xuất vật tư
- Thực thi sản xuất

Chuyển nhà máy
VIETBIKE - HANOI

AD **admin**
admin@example.com

Ngôn ngữ
vi

Đăng xuất
Hu gọn thanh bên

Người dùng

Quản lý người dùng backend theo đúng các trường tài khoản và hành động vòng đời đã công bố.

+ Tạo người dùng

Tên đăng nhập	Vai trò	Trạng thái
kiemtoan.tamthoi kiemtoan.tamthoi@vietbike.vn	OPERATOR	ACTIVE
quanly.saigon quanly.saigon@vietbike.vn	OPERATOR	ACTIVE
kehoach.vietbike kehoach.vietbike@vietbike.vn	OPERATOR	ACTIVE
congngan.hanoi congngan.hanoi@vietbike.vn	OPERATOR	ACTIVE
quanly.hanoi quanly.hanoi@vietbike.vn	OPERATOR	ACTIVE
admin admin@example.com	ADMIN	ACTIVE

Số dòng mỗi trang: 20

Hiển thị 1 đến 6 trong tổng số 6 kết quả Quay lại Tiếp

Figure 29. User management with account lifecycle status

4.2.21. Role Management

This screen defines the responsibility groups used by dynamic role-based access control. Active roles provide a reusable basis for assigning operational permissions to users.

OmniPlant

TỔNG QUAN

Bảng điều khiển

QUẢN TRỊ

Người dùng

Vai trò

Quyền hạn

Phạm vi truy cập

Gán quyền người dùng

LƯUỒNG SẢN XUẤT

Đơn bán hàng

Hoạch định vật tư

Lệnh sản xuất

Xuất vật tư

Thực thi sản xuất

Chuyển nhà máy

VIETBIKE - HANOI

AD admin
admin@example.com

Ngôn ngữ

vi

Đăng xuất

Như gọn thành bên

Vai trò

Quản lý vai trò và gán quyền hạn trong toàn hệ thống.

Tạo vai trò

Danh sách quyền của vai trò được đọc từ endpoint backend authoritative.

Mã	Tên	Loại	Trạng thái	
ADMIN	ADMIN System administrator with full access	Hệ thống	ACTIVE	...
MANAGER	MANAGER Production/inventory manager	Hệ thống	ACTIVE	...
KEHOACH	Nhân viên kế hoạch sản xuất Vai trò demo: chạy MRP, duyệt đề xuất cung ứng, đọc dữ liệu sản xuất	Tùy chỉnh	ACTIVE	...
OPERATOR	OPERATOR Shop floor operator with limited access	Hệ thống	ACTIVE	...

Số dòng mỗi trang: 20

Hiện thị 1 đến 4 trong tổng số 4 kết quả

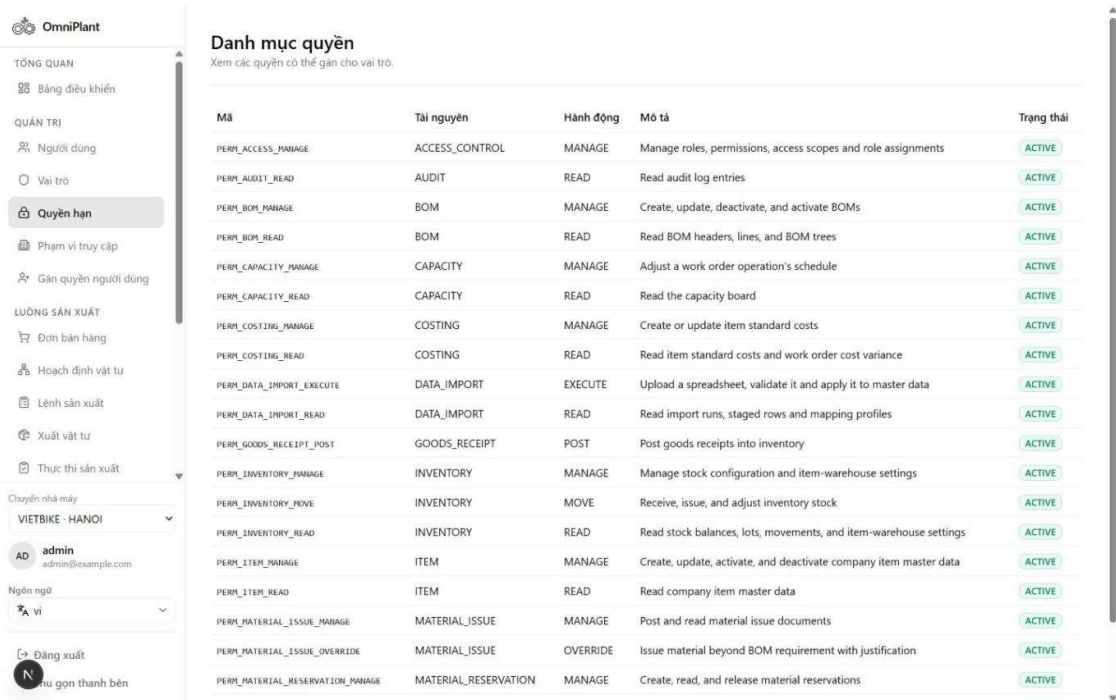
Quay lại

Tiếp

Figure 30. Role management with active responsibility groups

4.2.22. Permission Catalog

This screen lists the resource and action permissions recognized by the system. The catalog makes authorization rules explicit and supports consistent role configuration.



Mã	Tài nguyên	Hành động	Mô tả	Trạng thái
PERM_ACCESS_MANAGE	ACCESS_CONTROL	MANAGE	Manage roles, permissions, access scopes and role assignments	ACTIVE
PERM_AUDIT_READ	AUDIT	READ	Read audit log entries	ACTIVE
PERM_BOM_MANAGE	BOM	MANAGE	Create, update, deactivate, and activate BOMs	ACTIVE
PERM_BOM_READ	BOM	READ	Read BOM headers, lines, and BOM trees	ACTIVE
PERM_CAPACITY_MANAGE	CAPACITY	MANAGE	Adjust a work order operation's schedule	ACTIVE
PERM_CAPACITY_READ	CAPACITY	READ	Read the capacity board	ACTIVE
PERM_COSTING_MANAGE	COSTING	MANAGE	Create or update item standard costs	ACTIVE
PERM_COSTING_READ	COSTING	READ	Read item standard costs and work order cost variance	ACTIVE
PERM_DATA_IMPORT_EXECUTE	DATA_IMPORT	EXECUTE	Upload a spreadsheet, validate it and apply it to master data	ACTIVE
PERM_DATA_IMPORT_READ	DATA_IMPORT	READ	Read import runs, staged rows and mapping profiles	ACTIVE
PERM_GOODS_RECEIPT_POST	GOODS_RECEIPT	POST	Post goods receipts into inventory	ACTIVE
PERM_INVENTORY_MANAGE	INVENTORY	MANAGE	Manage stock configuration and item-warehouse settings	ACTIVE
PERM_INVENTORY_MOVE	INVENTORY	MOVE	Receive, issue, and adjust inventory stock	ACTIVE
PERM_INVENTORY_READ	INVENTORY	READ	Read stock balances, lots, movements, and item-warehouse settings	ACTIVE
PERM_ITEM_MANAGE	ITEM	MANAGE	Create, update, activate, and deactivate company item master data	ACTIVE
PERM_ITEM_READ	ITEM	READ	Read company item master data	ACTIVE
PERM_MATERIAL_ISSUE_MANAGE	MATERIAL_ISSUE	MANAGE	Post and read material issue documents	ACTIVE
PERM_MATERIAL_ISSUE_OVERRIDE	MATERIAL_ISSUE	OVERRIDE	Issue material beyond BOM requirement with justification	ACTIVE
PERM_MATERIAL_RESERVATION_MANAGE	MATERIAL_RESERVATION	MANAGE	Create, read, and release material reservations	ACTIVE

Figure 31. Permission catalog with resource and action codes

4.2.23. Access Scope Management

This screen defines where an assigned role is allowed to operate. Organization-level scopes restrict data and actions to the appropriate Company, Plant or Warehouse context.

OmniPlant

TỔNG QUAN

Bảng điều khiển

QUẢN TRỊ

Người dùng

Vai trò

Quyền hạn

Phạm vi truy cập

Gán quyền người dùng

LƯUỒNG SẢN XUẤT

Đơn bán hàng

Hoạch định vật tư

Lệnh sản xuất

Xuất vật tư

Thực thi sản xuất

Chuyển nhà máy

VIETBIKE - HANOI

AD admin
admin@example.com

Ngôn ngữ

vi

Đăng xuất

Như gọn thành bên

Phạm vi truy cập

Xác định tài nguyên tổ chức mà vai trò được phép truy cập.

+ Tạo phạm vi truy cập

Tài nguyên phạm vi chỉ có thể thêm. Không thể xóa riêng từng tài nguyên; hãy vô hiệu phạm vi hoặc thu hồi các gán quyền nếu thêm nhầm.

Mã	Tên	Loại	Trạng thái
GLOBAL_ALL	Global access System-wide access scope for platform administrators	GLOBAL	ACTIVE
PHAMVI-HANOI	Chi nhà máy Hà Nội Phạm vi dữ liệu demo	PLANT	ACTIVE
PHAMVI-KHOTP	Chi kho thành phẩm Hà Nội Phạm vi dữ liệu demo	WAREHOUSE	ACTIVE
PHAMVI-SAIGON	Chi nhà máy Sài Gòn Phạm vi dữ liệu demo	PLANT	ACTIVE
PHAMVI-TOANCONGTY	Toàn công ty Xe đạp Việt Phạm vi dữ liệu demo	COMPANY	ACTIVE

Số dòng mỗi trang 20

Hiện thị 1 đến 5 trong tổng số 5 kết quả Quay lại Tiếp

Figure 32. Access Scope management across organization levels

4.2.24. User Assignment

This screen links a user with a role and an access scope. The assignment form combines responsibility and organizational boundary into one authorization record.

OmniPlant

TỔNG QUAN

- Bảng điều khiển

QUẢN TRỊ

- Người dùng
- Vai trò
- Quyền hạn
- Phạm vi truy cập
- Gán quyền người dùng**

LƯUỒNG SẢN XUẤT

- Đơn bán hàng
- Hoạch định vật tư
- Lệnh sản xuất
- Xuất vật tư
- Thực thi sản xuất

Chuyển nhà máy: VIETBIKE - HANOI

AD **admin**
admin@example.com

Ngôn ngữ: vi

[Đăng xuất]
hu gọn thành bên

Gán quyền người dùng

Gán vai trò và phạm vi dữ liệu cho người dùng.

[+ Tạo gán quyền](#)

Tên đăng nhập	Vai trò	Phạm vi truy cập	Hết hạn	Trạng thái	
kiemtoan.tamthoi	OPERATOR	Chỉ kho thành phẩm Hà Nội	8/21/2026	ACTIVE	Thu hồi
quanly.saigon	MANAGER	Chỉ nhà máy Sài Gòn	—	ACTIVE	Thu hồi
kehoach.vietbike	Nhân viên kế hoạch sản xuất	Toàn công ty Xe đạp Việt	—	ACTIVE	Thu hồi
congnhan.hanoi	OPERATOR	Chỉ nhà máy Hà Nội	—	ACTIVE	Thu hồi
quanly.hanoi	MANAGER	Chỉ nhà máy Hà Nội	—	ACTIVE	Thu hồi
admin	ADMIN	Global access	—	ACTIVE	Thu hồi

Số dòng mỗi trang: 20

Hiện thị 1 đến 6 trong tổng số 6 kết quả

[Quay lại](#) [Tiếp](#)

Figure 33. User Assignment form linking User, Role and Scope

4.2.25. Organization Structure

This screen represents the Company, Plant and Warehouse hierarchy used throughout OmniPlant. The structure provides the organizational context for inventory, production and scoped authorization.

OmniPlant

Cơ cấu tổ chức

Công ty Cổ phần Xe đạp Việt ACTIVE

Mã công ty: VIETBIKE Tổng số nhà máy: 2

Nhà máy Kho

Q Tim kiếm + Tạo nhà máy

ALL Đặt lại

Mã nhà máy	Tên nhà máy	Múi giờ	Trạng thái
HANOI	Nhà máy Hà Nội	UTC	ACTIVE
SAIGON	Nhà máy Sài Gòn	UTC	ACTIVE

Chuyển nhà máy: VIETBIKE - HANOI

admin admin@example.com

Ngôn ngữ: vi

Đăng xuất

Nhu gọn thành bên

Figure 34. Organization structure with Company, Plant and Warehouse evidence

4.2.26. Unit of Measure Management

This screen maintains reusable units of measure for item and transaction data. Standardized codes help keep quantities consistent across BOM, inventory and production workflows.

OmniPlant

Vị trí truy cập

Gán quyền người dùng

Nhật ký kiểm toán

LƯU TRÌNH SẢN XUẤT

Đơn bán hàng

Hoạch định vật tư

Lệnh sản xuất

Xuất vật tư

Thực thi sản xuất

Nhập kho thành phẩm

Danh mục vật tư

Đơn vị tính

Kho

Lô tồn kho

Biến động kho

Định mức nguyên vật...

Chuyển nhà máy

VIETBIKE - SAIGON

AD admin
admin@example.com

Ngôn ngữ

Việt

Đăng xuất

Nhau gọn thanh bên

Đơn vị tính

Tổng 7 đơn vị

+ Thêm đơn vị

Đặt lại

Tìm theo mã hoặc tên...

ALL

Mã	Tên	Mô tả	Trạng thái	Thao tác
B0	Bộ	Nhóm chi tiết đi liền nhau	Hoạt động	...
CAI	Cái	Đơn vị đếm từng chiếc	Hoạt động	...
GIO	Giờ công	Thời gian lao động	Hoạt động	...
KG	Kilôgam	Khối lượng	Hoạt động	...
LIT	Lít	Thể tích	Hoạt động	...
MET	Mét	Chiều dài	Hoạt động	...
TA	Tạ	Đơn vị cũ, không còn dùng	Ngưng hoạt động	...

Số dòng mỗi trang 20

Hiện thị 1 đến 7 trong tổng số 7 kết quả

Quay lại

Tiếp

Figure 35. UOM management with reusable measurement codes

4.2.27. Audit Logs

This screen provides evidence of sensitive system and data changes. Actor, entity, action and timestamp fields support accountability and operational review.

Thời gian	Người dùng	Hành động	Mô-đun	Đối tượng	Trạng thái	Mã truy vết
16:26 17 thg 8, 2026	admin	ITEM_DEACTIVATED	Item	Vành hộp kim 24 inch (ngừng dùng)	SUCCESS	d2e167a257a34e44
15:57 17 thg 8, 2026	admin	ITEM_ACTIVATED	Item	—	SUCCESS	d56402b089944d
23:53 14 thg 8, 2026	admin	GOODS_RECEIPT_POSTED	GoodsReceipt	—	SUCCESS	a14f250b1ee5468a
23:53 14 thg 8, 2026	admin	PURCHASE_ORDER_SENT	PurchaseOrder	—	SUCCESS	d87c44ee3b45480c
23:53 14 thg 8, 2026	admin	GOODS_RECEIPT_CANCELLED	GoodsReceipt	—	SUCCESS	b083436fc65d425c
23:53 14 thg 8, 2026	admin	GOODS_RECEIPT_POSTED	GoodsReceipt	—	SUCCESS	a6a32ec27f1f4813
23:53 14 thg 8, 2026	admin	GOODS_RECEIPT_POSTED	GoodsReceipt	—	SUCCESS	4661f2c693274e87
23:53 14 thg 8, 2026	admin	PURCHASE_ORDER_SENT	PurchaseOrder	—	SUCCESS	f08098284f04ac7
23:53 14 thg 8, 2026	admin	PURCHASE_ORDER_CREATED	PurchaseOrder	—	SUCCESS	c9520e270b1e45ad
23:53 14 thg 8, 2026	admin	PURCHASE_ORDER_CANCELLED	PurchaseOrder	—	SUCCESS	6a58c166d6c0447d
23:53 14 thg 8, 2026	admin	PURCHASE_ORDER_CREATED	PurchaseOrder	—	SUCCESS	8d28e1ee73ce4fc8
23:53 14 thg 8, 2026	admin	PURCHASE_ORDER_SENT	PurchaseOrder	—	SUCCESS	d9f589e873c040d3
23:53 14 thg 8, 2026	admin	PURCHASE_ORDER_CREATED	PurchaseOrder	—	SUCCESS	3694980d68e4915
23:53 14 thg 8, 2026	admin	PURCHASE_ORDER_CREATED	PurchaseOrder	—	SUCCESS	83a0d943c089c407c
23:53 14 thg 8, 2026	admin	PURCHASE_REQUISITION_CONVERTED	PurchaseOrder	—	SUCCESS	c5a8e16e2b0744c6
23:53 14 thg 8, 2026	admin	PURCHASE_REQUISITION_APPROVED	PurchaseRequisition	—	SUCCESS	230bc2d4d57441e5
23:53 14 thg 8, 2026	admin	PURCHASE_REQUISITION_CANCELLED	PurchaseRequisition	—	SUCCESS	958451c2f5b04f9d
23:53 14 thg 8, 2026	admin	PURCHASE_REQUISITION_CREATED	PurchaseRequisition	—	SUCCESS	805e2609e8d4311
23:53 14 thg 8, 2026	admin	PURCHASE_REQUISITION_REJECTED	PurchaseRequisition	—	SUCCESS	b793c8d614ce4a3a
23:53 14 thg 8, 2026	admin	PURCHASE_REQUISITION_CREATED	PurchaseRequisition	—	SUCCESS	922809f93164846

Figure 36. Audit Logs with actor, entity, action and timestamp evidence

4.3. Validation and Acceptance Results

The implementation was validated through focused domain tests, API contract tests, component presentation tests, permission/navigation tests and live browser acceptance. The latest complete repository gate recorded before this report included 59 Vitest files with 265 passing tests, strict TypeScript, EN/VI translation parity, API coverage checks, production build, ESLint and diff validation. Earlier Scope 4 acceptance specifically covered planning, allocation, reservation, issue, WIP, partial receipt, rejection, QC, stock effects and cancellation.

Gate	Recorded outcome
Automated regression	59 Vitest files / 265 tests passed at the completed integration gate
Type safety	Strict TypeScript and runtime schema/adaptor validation
API coverage	Guard prevents MVP routes from silently reverting to mock-only runtime boundaries
Internationalization	English/Vietnamese key parity
Responsive QA	Desktop and tablet checks with no document-level horizontal overflow

Gate	Recorded outcome
RBAC	Admin, Manager and Operator permissions/actions verified; no-permission state covered by regression
Inventory effects	Reservation, issue, HOLD receipt, QC AVAILABLE/REJECTED and Dashboard invalidation verified

Table 9. Quality and acceptance evidence

4.4. Operational Demonstration Result

The presentation fixture demonstrated a completed Planning Run containing two demands, eight requirement lines, six shortages, three MAKE suggestions and three BUY recommendations. Work Orders were exercised across PLANNED, RELEASED, BLOCKED, IN_PROGRESS and COMPLETED states. Execution evidence recorded good, scrap and rework quantities, while Production Receipts covered DRAFT, PENDING_APPROVAL, APPROVED/HOLD, QC AVAILABLE and QC REJECTED outcomes.

The demonstration also showed partial Sales Order fulfillment after QC AVAILABLE output, confirming that accepted finished goods flow back into business demand status rather than appearing only in an isolated inventory screen.

4.5. Engineering Discussion

4.5.1. Planning Run versus Standalone Production Estimation

Both functions use BOM explosion and stock comparison, but their business purposes differ. A standalone estimator answers an ad-hoc question about one product and target quantity. The Planning Run begins with confirmed demand, creates an auditable immutable snapshot, applies safety-stock netting, produces supply suggestions and continues into Work Orders. For this reason, the Planning Run is the authoritative Capstone 2 evidence and Production Estimation remains outside the focused scope.

4.5.2. Immutable Snapshots and Idempotency

Manufacturing records must remain explainable after engineering masters change. Snapshotting BOM and Routing content into the Work Order provides that stability. Idempotency addresses a different time dimension: it prevents a repeated network submission from creating a second logical document. Together, these mechanisms protect both historical meaning and transactional uniqueness.

4.5.3. Availability and Quality Semantics

On-hand quantity alone is insufficient for planning. OmniPlant separates reservations and quality status so that only eligible stock becomes projected supply. The same rule is applied across Dashboard alerts, MRP, reservation, issue selection and Sales Order fulfillment. This consistency is more important than any single screen because a mismatch would produce false production readiness.

4.5.4. ERP Execution without MES

Capstone 2 records actual production manually rather than simulating machine integration. The interface captures quantities, times, actor and notes, while the model preserves an extensible execution-source boundary. This approach delivers a usable ERP process now and avoids misleading claims about real-time telemetry, OEE or automated shop-floor dispatching.

4.6. Known Limitations

- Full BUY-side execution is deferred; a BUY suggestion does not create a committed scheduled receipt in the focused MVP.
- Capacity is advisory and retained as a supporting master/scheduling seam; automatic finite optimization is not implemented.
- The current three-actor model assigns Production Receipt approval and QC disposition to Manager through separate permissions; a dedicated QC role can be added later.
- Detailed manufacturing cost, labor/overhead valuation, OEE and profitability are not calculated.
- MES, PLC and IoT data are not integrated; all execution confirmation is manual.
- Presentation fixture audits documented idempotency/atomicity and availability edge cases for continued backend hardening; the business contracts remain explicitly tested and monitored.

CHAPTER 5. CONCLUSION AND FUTURE WORKS

5.1. Conclusion

Capstone 2 converted OmniPlant from an analysis-and-prototype project into an integrated Manufacturing ERP application with a coherent operational spine. The system captures customer demand, performs material planning, creates and controls Work Orders, preserves inventory and lot traceability, records manual production execution, approves output and enforces a quality gate before finished goods become available.

The implementation demonstrates that a focused SME-oriented ERP can maintain enterprise-level safeguards without adopting the full complexity of a large commercial suite. Dynamic permissions, Plant scoping, immutable snapshots, append-only inventory movements, idempotent commands, explicit states and responsive list/form workspaces provide a disciplined foundation for real operational use and later expansion.

The most important project result is the continuity of evidence. A reviewer can navigate from a Sales Order line through an immutable Planning Run and Work Order to issued material lots, WIP confirmation, Production Receipt, QC outcome and output inventory. This continuity is the defining achievement of the Capstone 2 scope.

5.2. Future Works

- **Purchasing:** Implement suppliers, Purchase Requests/Requisitions, Purchase Orders, Goods Receipt, approval and incoming-quality workflow; only confirmed receipts should become scheduled supply.
- **Sales fulfillment:** Add delivery allocation, goods issue, invoicing, payment and customer-credit controls.
- **Manufacturing costing:** Add material, labor and overhead rates, WIP valuation, cost variance and profitability analysis.
- **MES integration:** Connect ISA-95-aligned messages, PLC/IoT telemetry, automatic execution events, downtime and OEE while retaining manual fallback.
- **Advanced scheduling:** Add finite-capacity optimization, dispatching, rescheduling and scenario comparison beyond advisory overload warnings.
- **Quality management:** Add dedicated QC roles, inspection plans, measurements, non-conformance, quarantine and corrective actions.
- **Optional what-if analysis:** Migrate standalone Production Estimation to the backend only if Product Owner or presentation needs justify a separate non-executable calculator.

5.3. Recommended Capstone 3 Sequence

Priority	Extension	Reason
1	Purchasing and scheduled receipts	Closes the BUY suggestion loop and improves MRP realism.
2	Delivery and invoicing	Closes Sales Order fulfillment beyond production completion.
3	Costing and variance	Uses the quantity/time evidence already collected in Capstone 2.
4	Dedicated quality management	Expands the current HOLD/AVAILABLE/REJECTED gate into measurable inspection.
5	MES/OEE and finite scheduling	Builds on stable ERP documents after transactional foundations are proven.

Table 10. Recommended Capstone 3 implementation sequence

5.4. Operational State Reference

Document	Primary transition
Sales Order	DRAFT -> CONFIRMED -> IN_PRODUCTION -> PARTIALLY_FULFILLED -> FULFILLED -> CLOSED
Planning Run	PENDING/RUNNING -> COMPLETED or FAILED; result is immutable
Supply Suggestion	DRAFT -> APPROVED/REJECTED -> CONVERTED for eligible MAKE
Work Order	DRAFT -> PLANNED -> RELEASED -> IN_PROGRESS -> COMPLETED -> CLOSED; BLOCKED/CANCELLED branches
Production Receipt	DRAFT -> PENDING_APPROVAL -> APPROVED/HOLD -> AVAILABLE or REJECTED
Inventory Lot	AVAILABLE, HOLD or REJECTED; only AVAILABLE is eligible supply

Table 11. Primary document state transitions

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APPENDIX

A.1. Focused MVP Screen Inventory

Navigation group	Visible MVP workspaces
Overview	Dashboard
Admin	Users; Roles; Permissions; Access Scopes; User Assignments
Demand and planning	Sales Orders; Material Planning; Planning Run Detail
Production	Work Orders; Material Issues; Production Execution; Production Receipts
Manufacturing foundation	Item Master; UOM; Stock; Inventory Lots; Stock Movements; BOM

Table 12. Focused Capstone 2 MVP navigation

A.2. End-to-End Demonstration Checklist

1. Authenticate as Manager and select an authorized Plant.
2. Verify active Item, UOM, Warehouse, BOM and eligible inventory data.
3. Create and confirm a Sales Order with positive quantity and due date.
4. Run Material Planning and inspect requirements, shortage messages and MAKE/BUY suggestions.
5. Approve and convert an eligible MAKE suggestion into a Work Order.
6. Reserve 100% of required components and release the Work Order.
7. Post Material Issue using eligible lot/serial stock and confirm the movement/trace.
8. Record manual execution good, scrap, rework and actual time.
9. Create and submit a Production Receipt; approve it into HOLD.
10. Perform QC disposition and verify AVAILABLE or REJECTED output behavior.
11. Verify Sales Order fulfillment, inventory lot, stock movement, Dashboard and genealogy links.